

Minds are like
parachutes – they only
function when **OPEN**.

THOMAS DEWAR

Life is short.
Smile while
you still
have teeth.

The only impossible
journey is the one you
never begin.



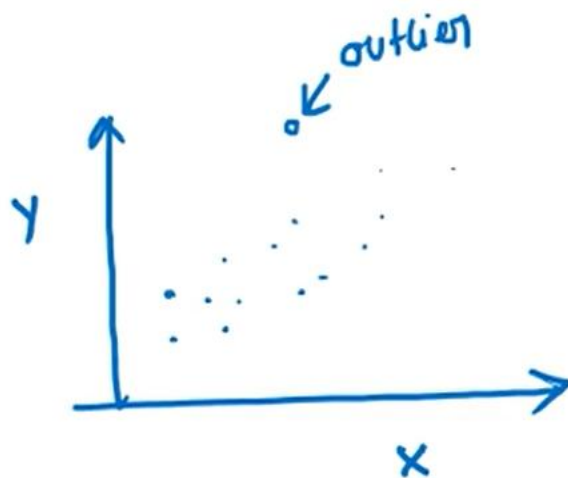
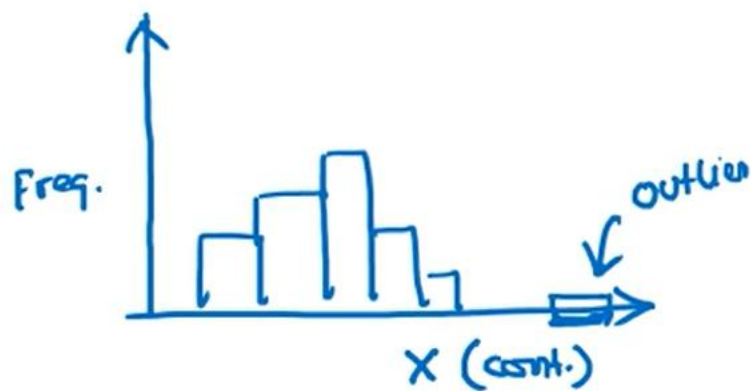
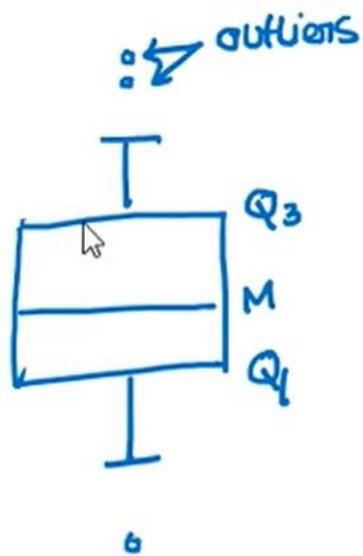
Module 4:- Anomaly Detection

Module 4:-Anomaly Detection

- 4.1 Outliers,
Causes of Outliers,
Anomaly detection techniques,
Outlier Detection using Statistics
- 4.2 Outlier Detection using Distance based method,
Outlier detection using density-based methods,
SMOTE

WHAT IS AN OUTLIER?

- In [statistics](#), an **outlier** is an observation point that is distant from other observations. (Wikipedia)



Outliers are important because:

- 1.They help **identify data entry or measurement errors**.
- 2.They reveal **rare or abnormal events such as fraud or attacks**.
- 3.They significantly **affect statistical measures like mean and standard deviation**.
- 4.They provide **valuable insights and improve model accuracy** when properly handled.

Outliers are important because:

They help identify data entry or measurement errors.

Detect Errors in Data Outliers often indicate mistakes in data collection or entry.

Example

Suppose the ages of students are:

18, 19, 20, 21, 200

Here 200 is likely a data entry error.

By detecting outliers we can:

correct the error

remove invalid values

Importance: improves data quality and accuracy.

Outliers are important because:

Identify Rare but Important Events

Outliers sometimes represent rare but meaningful observations.

Example

In credit card transactions:

Normal purchase: ₹500–₹5000

Transaction: ₹1,00,000

This outlier could indicate fraud.

Outliers are important because:

Affect Statistical Analysis

Outliers can significantly change statistical measures.

Example

Data: 10, 12, 13, 14, 100

Mean: $\text{Mean} = 29.8$

But the real values are around 12–14. So the outlier distorts the mean.

Measures affected:

Mean

Standard deviation

Regression models

Importance: helps choose robust statistical methods.

Outliers are important because:

Provide Insight into the Data

Outliers may reveal new patterns or unknown phenomena.

Example

A scientist studying temperature finds an extreme value.

This may indicate:

climate anomaly

sensor malfunction

new scientific observation

In business:

unusually high sales → successful campaign

Importance: leads to new discoveries or insights.

Outliers are important because:

Improve Machine Learning Models

Outliers can reduce model accuracy.

Example:

Linear regression line may shift due to extreme values.

Handling outliers improves:

prediction accuracy

model stability

training performance

Common handling methods:

removal

transformation

robust algorithms

Outliers

What are the types of Outliers?

Outlier can be of two types: **Univariate** and **Multivariate**.

Univariate outliers can be found when we look at distribution of a single variable.

Multi-variate outliers are outliers in an n -dimensional space. In order to find them, you have to look at distributions in multi-dimensions.

A **univariate outlier** is an unusual value detected by analyzing **one variable at a time**.

We only look at the **distribution of a single feature**.

Example

Consider students' marks:

Here **150** is a **univariate outlier** because it is far from the rest of the values when we examine **only the Marks variable**.

Methods to Detect Univariate Outliers

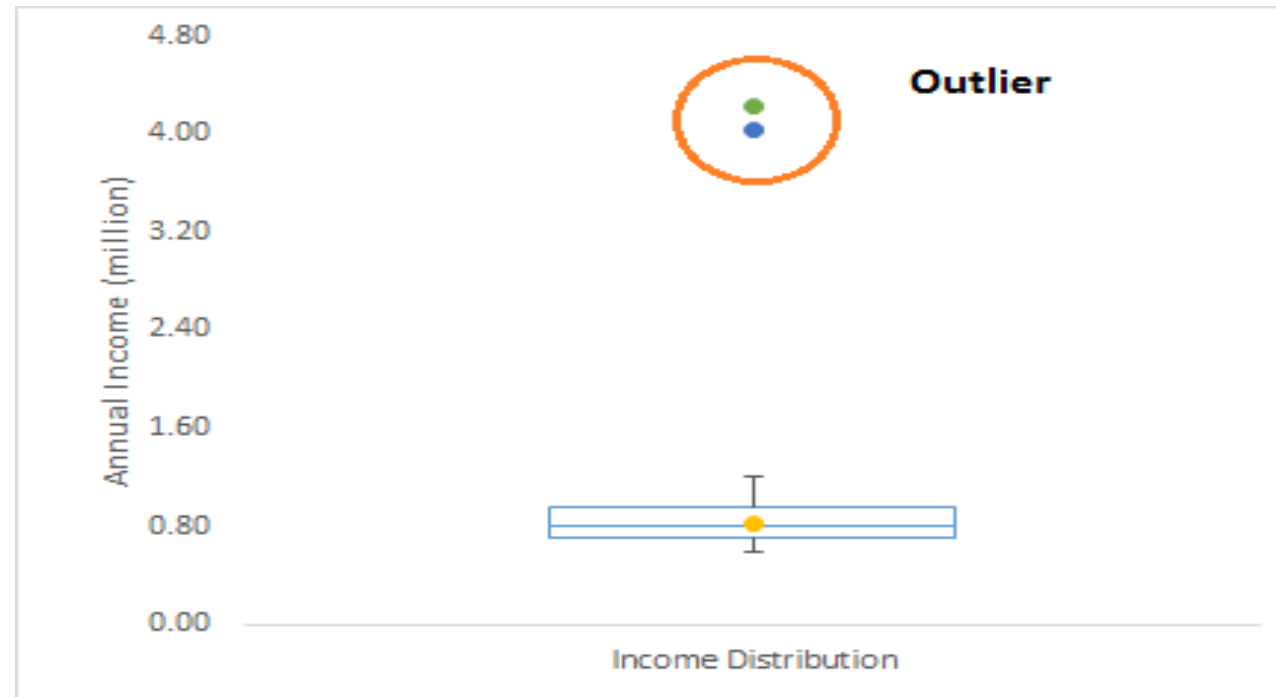
- Box Plot
- Z-score
- IQR method
- Histogram

Student	Marks
A	70
B	75
C	72
D	74
E	150

Key Idea

We check **one dimension (one feature)**.

Univariate outliers can be found when we look at distribution of a single variable. Let's take an example, we do customer profiling and find out that the average annual income of customers is \$0.8 million. But, there are two customers having annual income of \$4 and \$4.2 million. These two customers annual income is much higher than rest of the population. These two observations will be seen as Outliers.



Multivariate Outlier

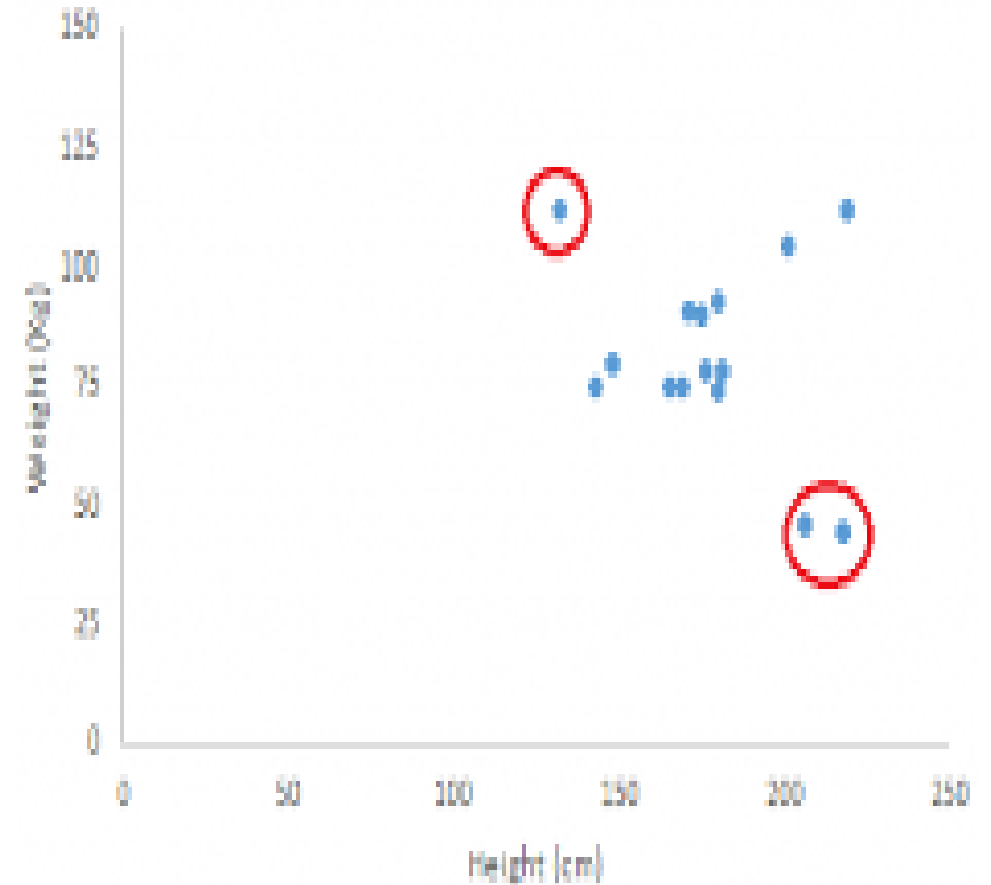
A **multivariate outlier** is an observation that looks normal in each individual variable but becomes unusual when **multiple variables are considered together**.

It occurs in **multi-dimensional space (n-dimensional space)**.

Example

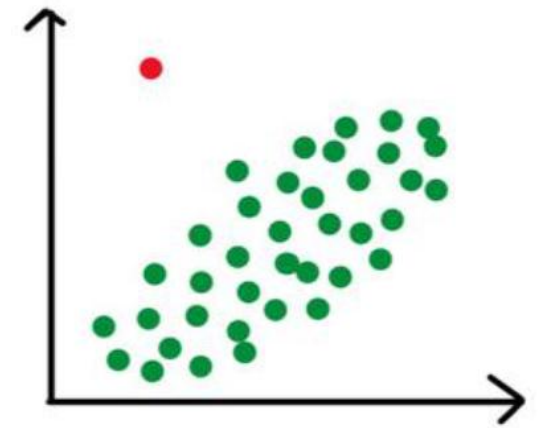
Suppose we analyze **Height and Weight**:

Person	Height (cm)	Weight (kg)
A	170	65
B	168	63
C	172	67
D	169	66
E	170	110

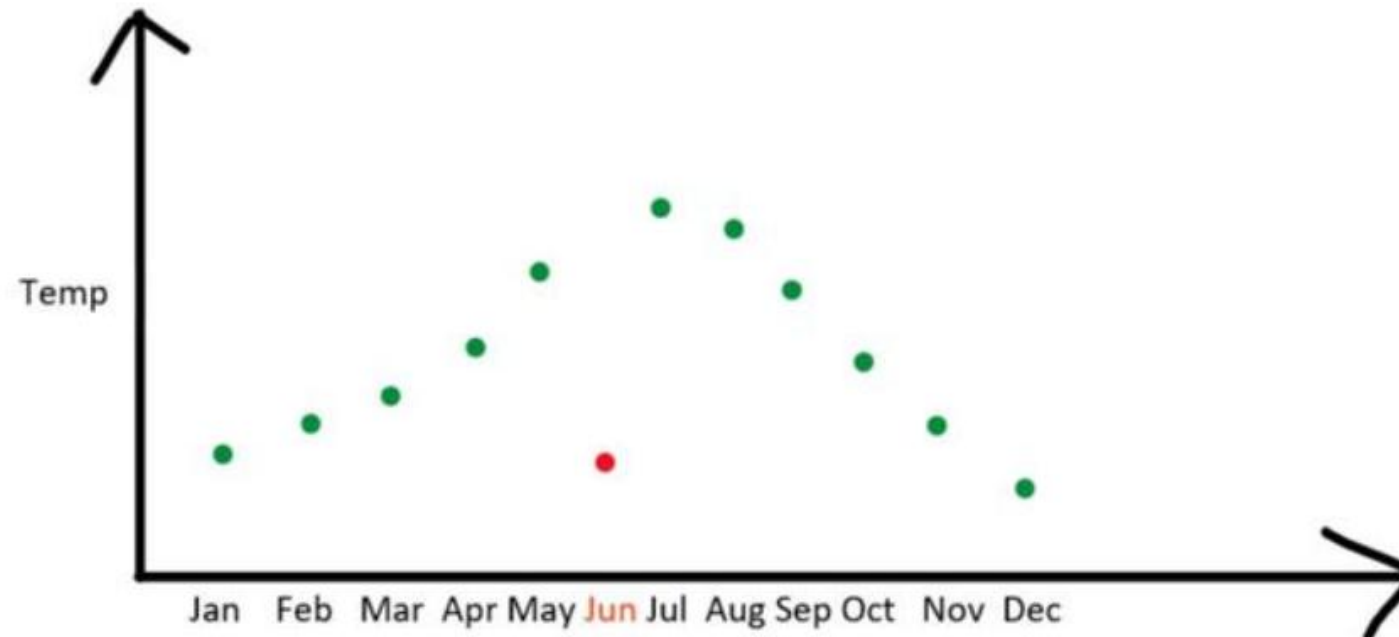


Types of Outliers (I)

- Three kinds: *global*, *contextual* and *collective* outliers
- **Global outlier** (or point anomaly)
 - Object is O_g if it significantly deviates from the rest of t
 - Ex. Intrusion detection in computer networks
 - Issue: Find an appropriate measurement of deviation
- **Contextual outlier** (or *conditional outlier*)



The red data point is a global outlier.



on a selected context
summer or winter?)
into two groups
t, e.g., time & location
the object, used in outlier
utliers—whose density
I context?

1. Global (Point) Outlier

Definition

A **global outlier** is a **single data point** that is **far away from the rest of the dataset**.

It looks unusual when compared with **all other values**.

Example

Marks of students:

Student	Marks
A	65
B	70
C	72
D	68
E	95

Here **95** is much higher than the others.

✔ 95 is a global outlier.

“One value is far away from all other values.”

2. Contextual Outlier

Definition

A **contextual outlier** is a value that is unusual **only within a specific context** (such as time, location, or condition).

The same value might be **normal in another context**.

Example

Temperature data:

Month	Temperature
April	35°C
May	38°C
June	40°C
December	35°C

35°C in **summer** → normal

35°C in **December** → unusual

✔ So 35°C in December is a contextual outlier.

Value is abnormal only in a specific situation

Types of Outliers (II)



Outlier

- **Collective Outliers**

- A subset of data objects *collectively* deviate significantly from the whole data set, even if the individual data objects may not be outliers
- Applications: E.g., *intrusion detection*:
 - When a number of computers keep sending denial-of-service packages to each other
- Detection of collective outliers
 - Consider not only behavior of individual objects, but also that of groups of objects
 - Need to have the background knowledge on the relationship among data objects, such as a distance or similarity measure on objects.
- A data set may have multiple types of outlier
- One object may belong to more than one type of outlier



3. Collective Outlier

Definition

A **collective outlier** occurs when a **group of data points together behaves abnormally**, even if individual points appear normal.

Example

Normal website traffic per minute:

100, 105, 110, 98, 102

During attack:

500, 520, 540, 510

Each value alone may seem possible, but **the entire sequence is abnormal**.

✓ This group forms a **collective outlier**.

A group of values together looks abnormal

Marks of students:

60, 62, 59, 61, 58, **120**

Question:

Identify the outlier and its type.

Solution

Most marks are between **58–62**.

120 is far away from the rest.

Temperature data:

Month	Temperature
January	20°C
February	22°C
March	25°C
April	30°C
January	35°C

Website traffic per minute:

100, 105, 98, 102, **450, 460, 470**

A student scores **100** marks in a class where most students score **70–80**.

Feature	Noise	Outlier	Anomaly
Meaning	Random error in data	Extreme data point	Unusual behavior or pattern
Information value	Usually useless	May be useful	Often important
Cause	Measurement or system error	Natural variation or error	Rare events or abnormal activity
Detection	Data cleaning methods	Statistical methods	Machine learning models
Example	Sensor glitch	Salary = ₹10,00,000 among ₹50,000	Fraud transaction

Suppose you measure **height of students in a class.**

Student	Height (cm)
A	165
B	168
C	170
D	169
E	205

What students think

205 cm is strange.

Explain

Outlier

- 205 cm is **far from other values**
- But it **may still be a real value**
- Maybe the student is a **basketball player**

Student	Height	Age
A	170	20
B	168	21
C	172	20
D	169	22
E	170	2

Now ask students:

Can a **2-year-old student** be in college?

No.

This becomes an **anomaly**.

Dataset: Employee salary

Employee	Salary
A	40,000
B	45,000
C	50,000
D	48,000
E	5,00,000

Explain:

Outlier

- CEO earning 5,00,000
- Rare but **valid**

Employee	Salary
A	40,000
B	45,000
C	-20000

Ask students:

Can salary be negative?

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Causes of Outliers

Whenever we come across outliers, the ideal way to tackle them is to **find out the reason of having these outliers.**

The **method to deal with them would then depend on the reason of their occurrence.**

Causes of outliers can be classified in two broad categories:

- 1. Artificial (Error) / Non-natural**
- 2. Natural.**

1. Artificial (Error / Non-Natural) Outliers

These occur because of mistakes in data collection, entry, measurement, or processing.

They do not represent the real situation, so usually they should be corrected or removed.

Causes of Outliers



- **Data Entry Errors:-**

- Human errors such as errors caused during data collection, recording, or entry can cause outliers in data.
- For example: Annual income of a customer is \$100,000. Accidentally, the data entry operator puts an additional zero in the figure. Now the income becomes \$1,000,000 which is 10 times higher.
- Evidently, this will be the outlier value when compared with rest of the population.

Causes of Outliers

- Data Entry Errors:-

You
deserve to
become the
person
you
dream of.

- Alexandra Elle

www.theerallivedin.com

Cod	Name	Age	ethnicity	Purchase value
1	Mark Grant	33	white	5500
2	Steve Manson	57	black	2500
3	Mary Jane	27	indigenous	3700
4	Antony Smith	470	white	2900
5	Aaron James	44	black	3300

Causes of Outliers

- **Measurement Error:**

- It is the most common source of outliers.

- This is caused when the measurement instrument used turns out to be **faulty**.

- For example: There are 10 weighing machines. 9 of them are correct, 1 is faulty.

Weight measured by people on the faulty machine will be higher / lower than the rest of people in the group. The weights measured on faulty machine can lead to outliers.

Causes of Outliers

- Measurement Error:



Causes of Outliers

Experimental Error:

Another cause of outliers is experimental error.

For example: In a 100m sprint of 7 runners, one runner missed out on concentrating on the 'Go' call which caused him to start late. Hence, this caused the runner's run time to be more than other runners. His total run time can be an outlier.



Causes of Outliers

- Intentional Outlier:
- This is commonly found in self-reported measures that involves sensitive data.
- For example:
- Teens would typically under report the **amount of alcohol that they consume**.
- Only a fraction of them would report actual value. Here actual values might look like outliers because rest of the teens are under reporting the consumption.

Causes of Outliers

- Data Processing Error:
- Whenever we perform data mining, we extract data from multiple sources.
- It is possible that some manipulation or extraction errors may lead to outliers in the dataset.



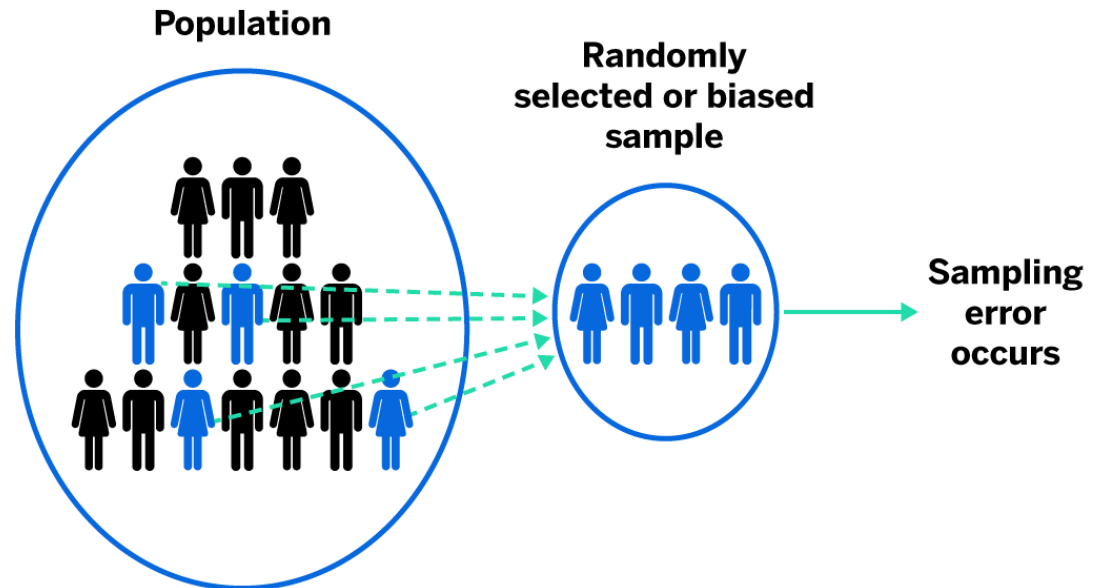
Causes of Outliers

Sampling error:

For instance, we have to measure the height of athletes.

By mistake, we include a few basketball players in the sample.

This inclusion is likely to cause outliers in the dataset.



1. Data Entry Error

Meaning

A mistake made while **typing** or recording data.

Example

Marks of students (out of 100):

Student	Marks
A	65
B	72
C	720
D	68

720 is impossible for an exam out of 100.

- ✓ **Cause:** typing error
- ✓ **Type:** Artificial outlier

2. Measurement Error

Meaning

Occurs when **measuring instruments** give incorrect readings.

Example

Temperature readings:

24°C, 25°C, 26°C, **120°C**, 25°C

A room cannot suddenly become **120°C**.

- ✓ **Cause:** faulty sensor
- ✓ **Type:** Artificial outlier

3. Experimental Error

Meaning

Errors occurring during **experiments** because of improper procedure or conditions.

Example

A chemistry experiment measures reaction time:

10 sec, 11 sec, 9 sec, **60 sec**

60 seconds may occur because the **equipment** was not set correctly.

- ✓ **Type:** Artificial outlier

4. Intentional Outlier

Meaning

When someone **intentionally** enters incorrect data.

Example

In an online survey:

Most ratings:

4, 5, 4, 5, 4

One person enters:

0

✓ The person intentionally gives an unrealistic value.

✓ **Type:** Artificial outlier

5. Data Processing Error

Meaning

Errors occurring during **data cleaning, transformation, or computation.**

Example

A system converts salary from rupees to thousands but makes a mistake.

Real salary:

₹50,000

Stored value:

500000

✓ **Cause:** wrong processing or calculation.

6. Sampling Error

Meaning

Occurs when the **sample** is not representative of the population.

Example

Suppose we want the **average height of college students**, but the sample includes a **child visiting campus**.

Person	Height
Student	170 cm
Student	168 cm
Student	172 cm
Visitor child	120 cm

120 cm appears as an outlier because of **incorrect sampling**.

✔ **Type:** Artificial outlier.

Natural Outliers

These occur because **extreme values genuinely exist in the real world**.

They are **valid observations** and should not always be removed.

1. Natural Variation

Meaning

Sometimes real-world data naturally has **extreme values**.

Example

Heights of students:

165, 168, 170, 169, **205**

205 cm is rare but possible.

✓ **Type:** Natural outlier.

2. Rare Events

Some events are **rare but real**.

Example

Daily rainfall (mm):

5, 6, 4, 7, **120**

120 mm may occur due to a **heavy storm**.

✓ **Natural outlier.**

3. Exceptional Performance

Extreme values caused by **high achievement or ability**.

Example

Employee sales (₹):

20,000

25,000

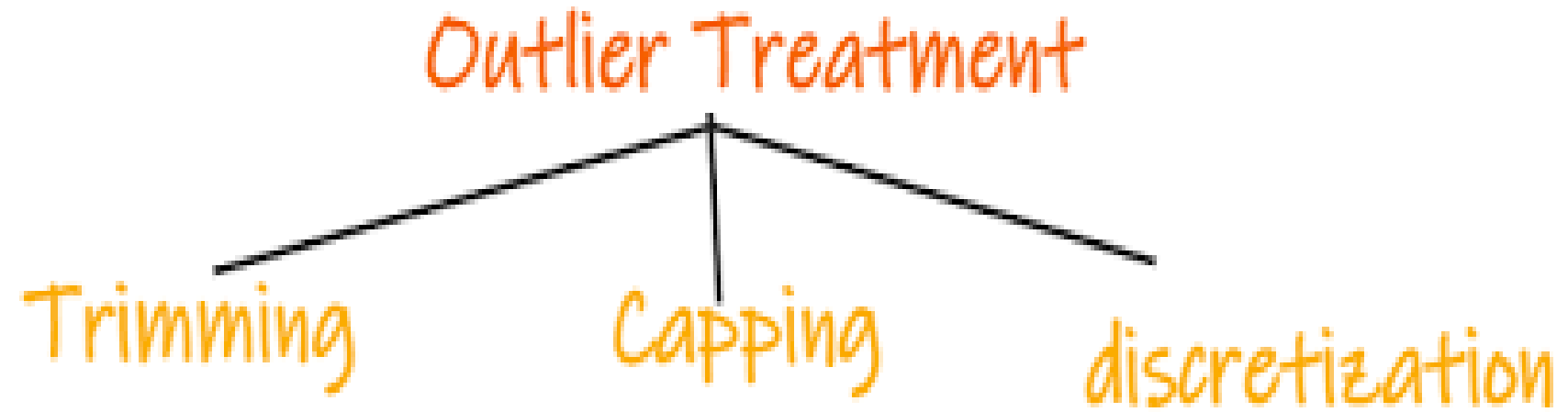
22,000

1,00,000

One salesperson performed exceptionally well.

✓ **Natural outlier.**

Handling the Outliers



Handling the Outliers

Suppose we have monthly salary data (₹):

Employee	Salary
A	40,000
B	42,000
C	45,000
D	43,000
E	2,00,000

The value 2,00,000 is an outlier.

Trimming = deleting the extreme value.

1. Trimming

Meaning

Trimming means removing the outlier from the dataset.

We simply delete extreme values.

After Trimming

Employee	Salary
A	40,000
B	42,000
C	45,000
D	43,000

The value 2,00,000 is removed.

When to Use

- When the outlier is caused by error
- Example: typing mistake

Handling the Outliers

Suppose we have monthly salary data (₹):

Employee	Salary
A	40,000
B	42,000
C	45,000
D	43,000
E	2,00,000

The value 2,00,000 is an outlier.

Capping = limiting extreme values.

2. Capping (Winsorization)

Meaning

Instead of removing the outlier, we replace it with a maximum acceptable value.

Example

Suppose the acceptable maximum salary is 50,000.

After Capping

Employee	Salary
A	40,000
B	42,000
C	45,000
D	43,000
E	50,000

The value 2,00,000 → replaced with 50,000.

When to Use

- When you do not want to lose data
- But want to reduce the effect of outliers

Handling the Outliers

Suppose we have monthly salary data (₹):

Employee	Salary
A	40,000
B	42,000
C	45,000
D	43,000
E	2,00,000

The value 2,00,000 is an outlier.

Discretization = converting numbers into groups.

3. Discretization

Meaning

Instead of keeping exact numbers, we convert numeric values into categories (bins).

Example

Salary categories:

Salary	Category
< 50,000	Low
50,000–1,00,000	Medium
> 1,00,000	High

After Discretization

Employee	Salary Category
A	Low
B	Low
C	Low
D	Low
E	High

Handling the Outliers

Imagine exam marks:

Marks

65

70

68

400

- **Trimming:** remove 400
- **Capping:** replace 400 with 100
- **Discretization:** convert to grade **A/B/C**

Outlier Removal / Detection Methods

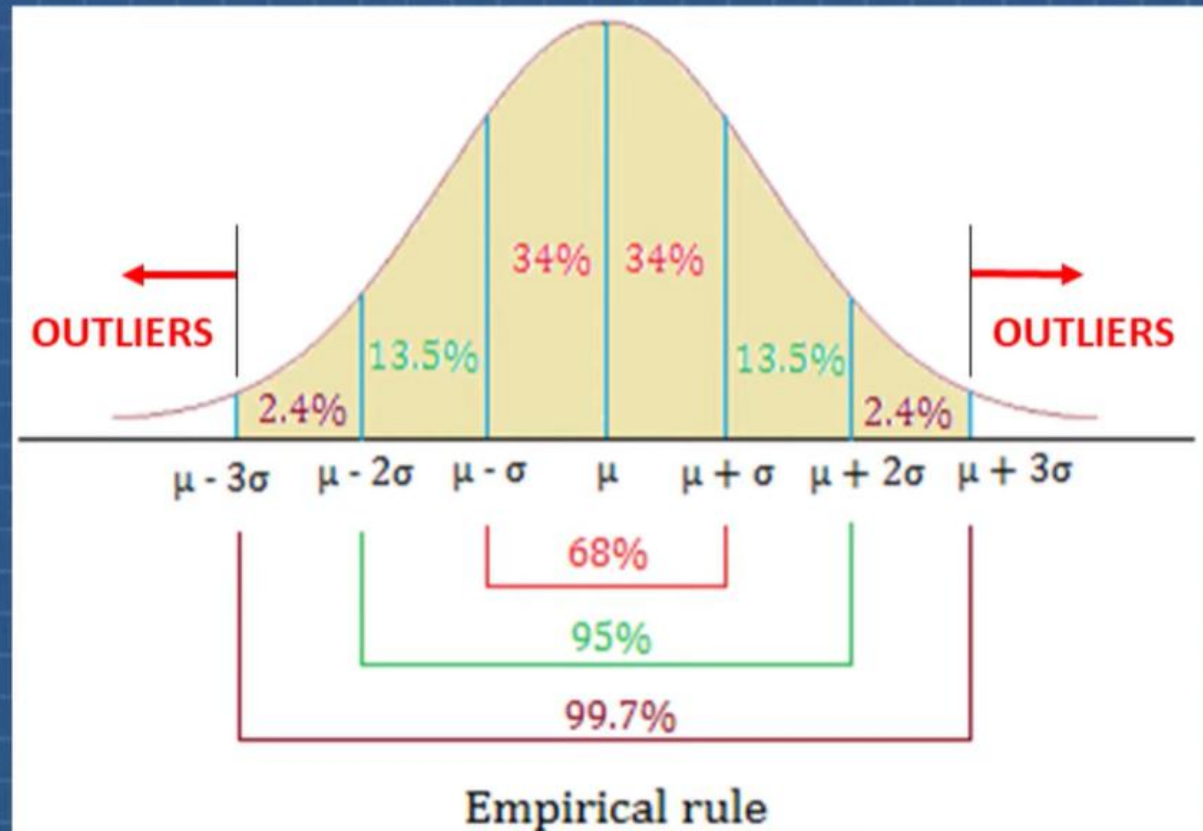
- STANDARD Z**

Exclude any values with Z score
< -3 (-2.5, -2)

OR

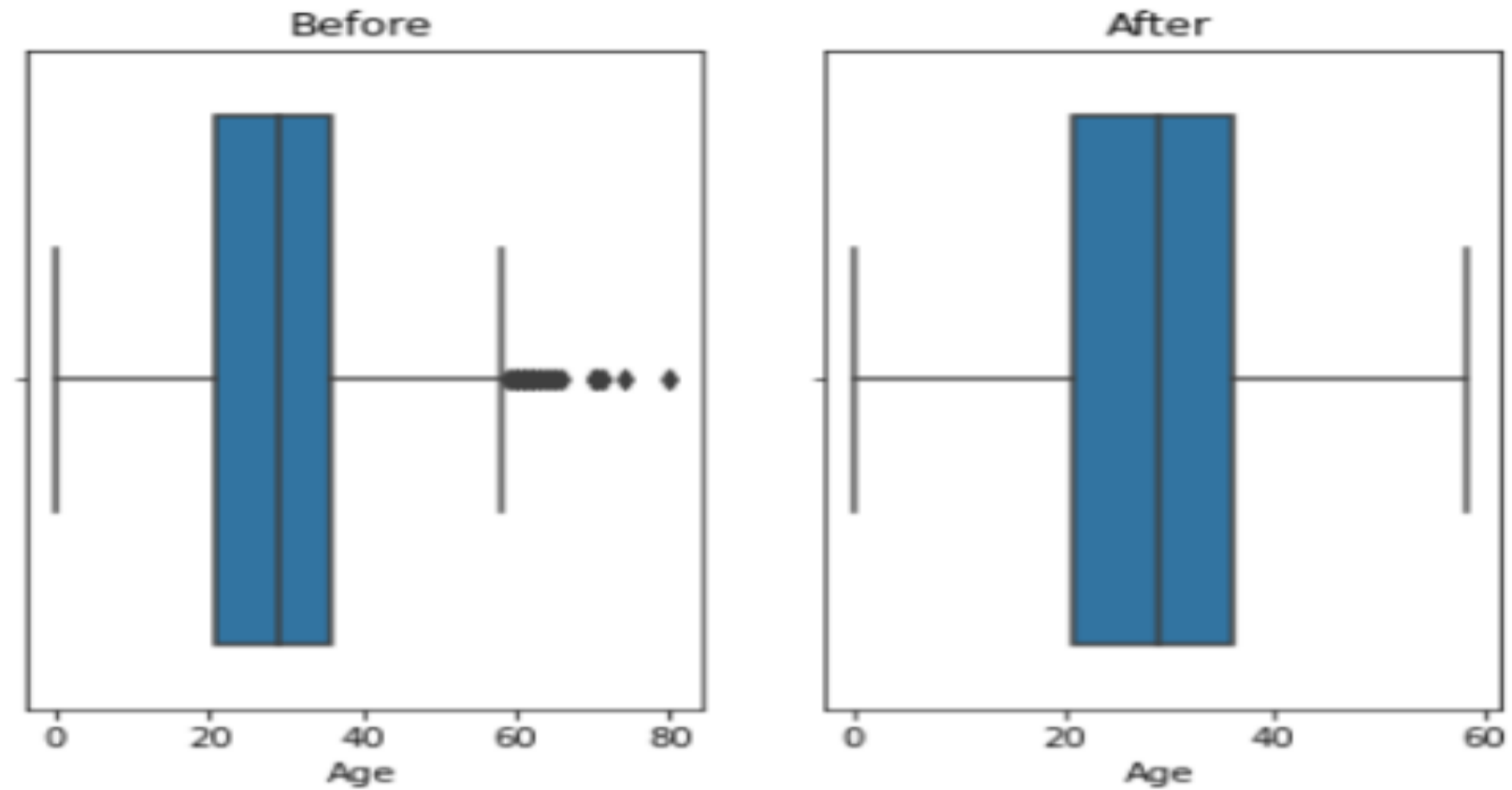
> 3 (2.5, 2)

$$z = \frac{x - \bar{x}}{s}$$



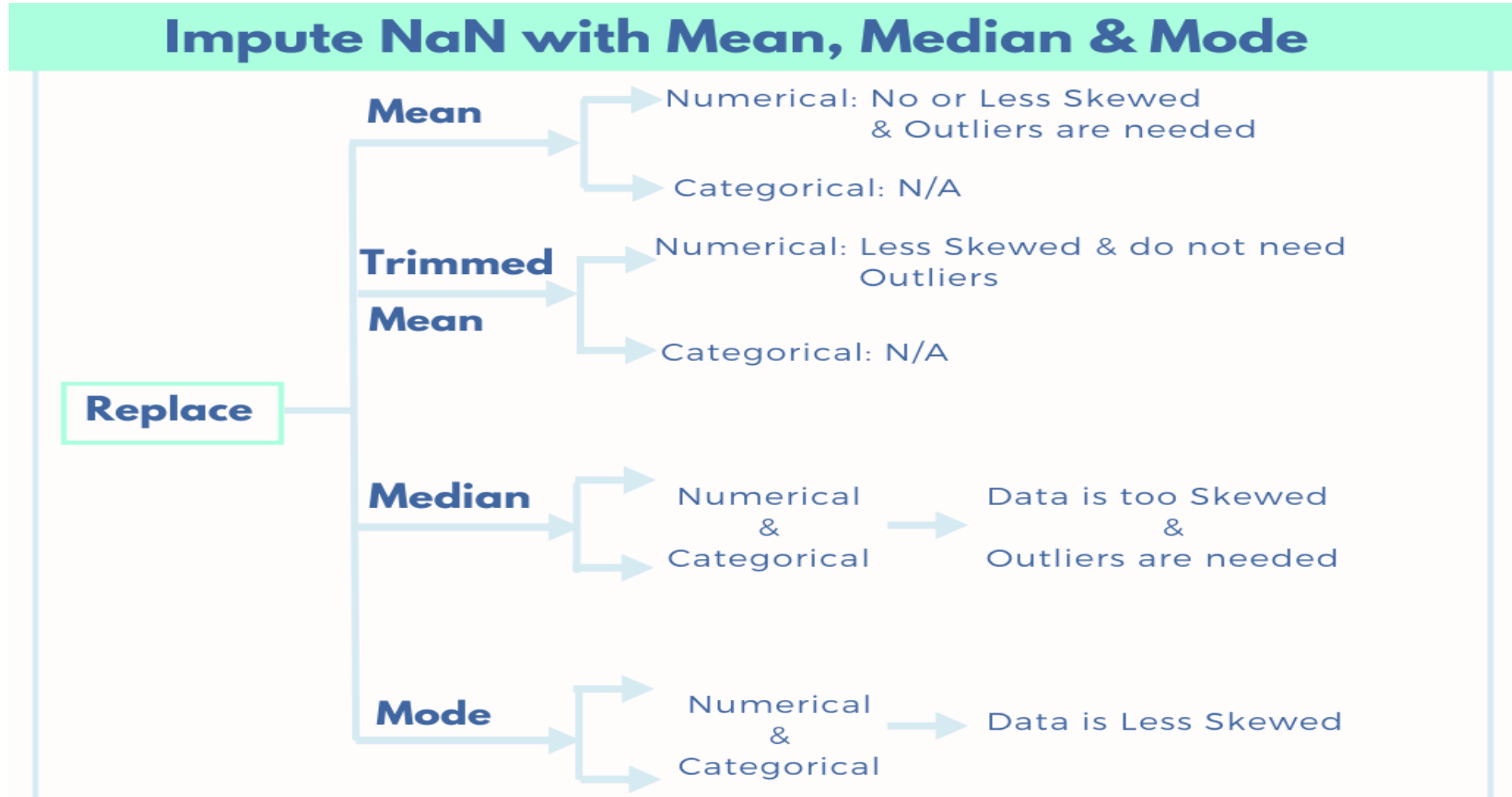
Handling the Outliers

Capping method



Handling the Outliers

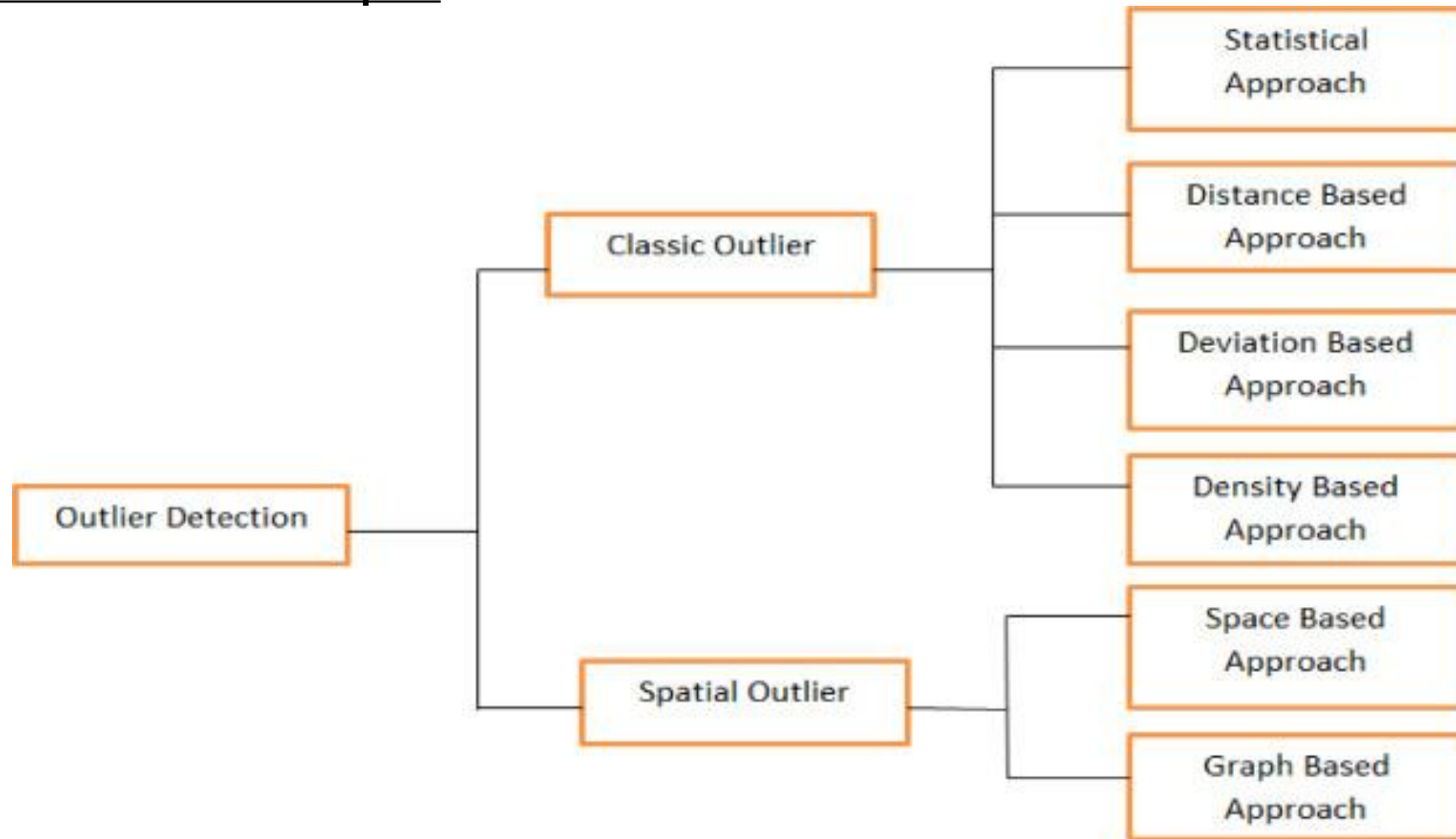
Mean/Median Imputation



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Anomaly detection techniques



- ▶ Find the outliers from the following data set.
10,20,30,40,500



- ▶ Find the outliers from the following data set.
10,206,240,300,350

Steps to find outliers

- ▶ Arrange the data in order from lowest to highest and find Q_1 and Q_3 .
✚
- ▶ Find the interquartile range $Q_3 - Q_1$.
- ▶ Multiply IQR by 1.5.
- ▶ Subtract step 3 from Q_1 and add in Q_3 .
- ▶ Check the data set for any data value that is smaller than $Q_1 - 1.5(\text{IQR})$ or larger than $Q_3 + 1.5(\text{IQR})$.

Example; find the outliers from the following data set.

10,11,15,25,35,30,7,68

..Arrange the data in order from lowest to highest and find Q1 and Q3.

Anomaly detection techniques

Sample Values

32, 37, 34, 35, 33, 35, 33, 32, 32, 4, 2, 55, 74, 102

5 Outliers Identified

2
4
55
74
102



**Not all of us can
do great things.
But we can do small things
with great love.**

-Mother Teresa

Z-Score based Outlier or Anomaly detection

- Suppose we have a dataset of daily sales revenue for a retail store over the past 30 days:
- [100, 150, 120, 125, 140, 130, 110, 135, 130, 150, 140, 100, 95, 80, 120, 125, 130, 100, 140, 135, 130, 145, 110, 120, 130, 135, 140, 125, 130, 120]

Z-Score based Outlier or Anomaly detection

Step 1: Calculate mean and standard deviation

- We first calculate the mean and standard deviation of the dataset:
- [100, 150, 120, 125, 140, 130, 110, 135, 130, 150, 140, 100, 95, 80, 120, 125, 130, 100, 140, 135, 130, 145, 110, 120, 130, 135, 140, 125, 130, 120]
- Mean: $\mu = \frac{\sum_{i=1}^{30} x_i}{n} = \frac{100+150+120+\dots+130+120}{30} = 123.5$

Z-Score based Outlier or Anomaly detection

Step 1: Calculate mean and standard deviation

- We first calculate the mean and standard deviation of the dataset:
- [100, 150, 120, 125, 140, 130, 110, 135, 130, 150, 140, 100, 95, 80, 120, 125, 130, 100, 140, 135, 130, 145, 110, 120, 130, 135, 140, 125, 130, 120]
- Mean: $\mu = \frac{\sum_{i=1}^{30} x_i}{n} = \frac{100+150+120+\dots+130+120}{30} = \underline{123.5}$
- *Standard deviation*:
- σ = $\sqrt{\frac{\sum_{i=1}^{30} (x_i - \mu)^2}{n-1}} = \sqrt{\frac{(100-123.5)^2 + (150-123.5)^2 + \dots + (120-123.5)^2}{30-1}} = 20.0$

Z-Score based Outlier or Anomaly detection

- **Step 2:** Calculate z-scores
- We then calculate the z-score for each data point, which represents how many standard deviations away from the mean the data point is:

- $$Z = \frac{x - \mu}{\sigma}$$

- **Step 2:** Calculate z-scores
- We then calculate the z-score for each data point, which represents how many standard deviations away from the mean the data point is:

- $$Z = \frac{x - \mu}{\sigma}$$

- $$Z_1 = \frac{100 - 123.5}{20} = -1.17$$

Z-Score based Outlier or Anomaly detection

- **Step 3:** Set a threshold
- We set a threshold for what we consider an anomaly.
- Usually z-score = 3 is considered as a cut-off value to set the limit.
which captures 99.7% of the data in a normal distribution
- Therefore, any z-score greater than +3 or less than -3 is considered as outlier which is pretty much similar to standard deviation method.

Z-Score based Outlier or Anomaly detection

- **Step 4:** Identify anomalies

< -3 . > 3

- [-1.17, 0.14, -0.28, -0.16, 0.56, 0.08, -0.89, 0.32, 0.08, 0.14, 0.56, -1.17, -1.45, -2.22, -0.28, -0.16, 0.08, -1.17, 0.56, 0.32, 0.08, 0.86, -0.89, -0.28, 0.08, 0.32, 0.56, -0.16, 0.08, -0.28]

No Outliers found in the given data

G15



fx

A

B

C

D

E

1

test score (from 100)

percentage

percentile rank

2

mohan

69

69%

100.00%

3

maria

56

56%

50.00%

4

sakib

45

45%

25.00%

5

tao

32

32%

12.50%

6

virat

27

27%

0.00%

7

khusbu

65

65%

75.00%

8

dmitry

61

61%

62.50%

9

selena

66

66%

87.50%

10

john

45

45%

25.00%

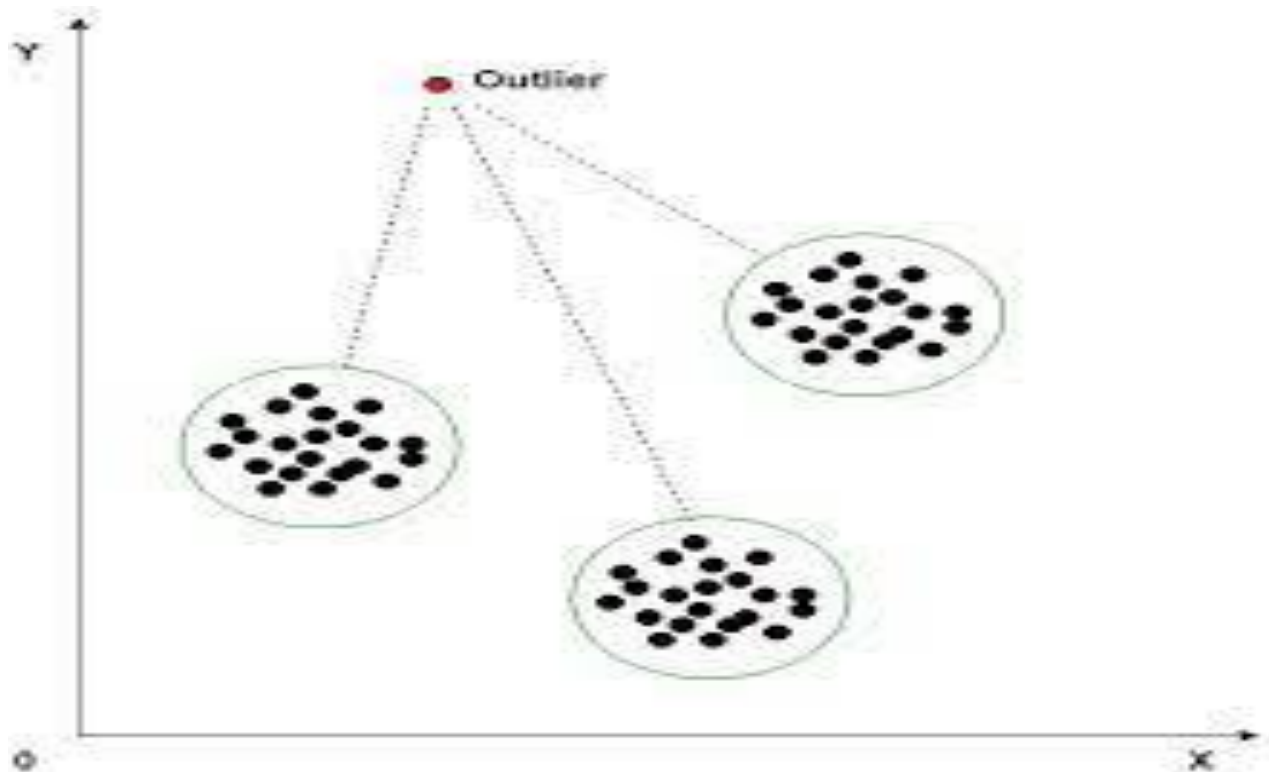
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Proximity-based Outlier Detection

- **Objects far away** from the others are outliers •
- The proximity of an outlier **deviates significantly from that of most of the others in the data set** •
- Distance-based outlier detection: An object **o** is an outlier if its **neighborhood does not have enough other points** •
- Density-based outlier detection: An object **o** is an outlier if its **density is relatively much lower** than that of its neighbors

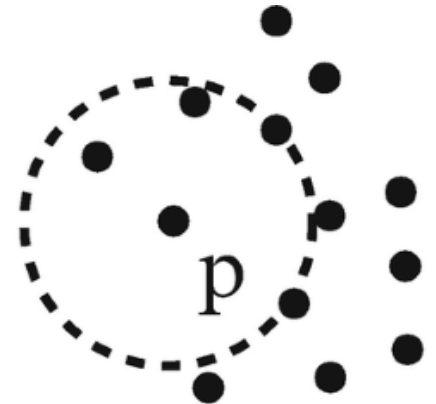
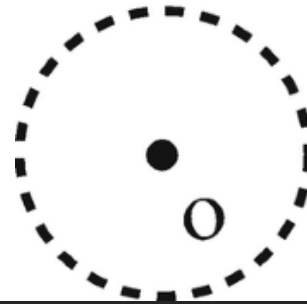
DISTANCE-BASED OUTLIER DETECTION

A distance-based outlier detection method consults the neighborhood of an object, which is defined by a **given radius**. An object is then considered **an outlier** if its **neighborhood does not have enough other points**. This is termed as Distance-Based Outlier Detection Methods.



DISTANCE-BASED OUTLIER DETECTION

- A $DB(p, D)$ -outlier is an object O in a dataset T such that at least a fraction p of the objects in T lie at a distance greater than distance D from O
- The larger D , the more outlying
- The larger p , the more outlying



DISTANCE-BASED OUTLIER DETECTION

KNN-based outlier detection works by measuring the distance to each point's K-nearest neighbors. Points with a **significantly larger average distance** to their neighbors are considered outliers

KNN-based outlier detection identifies anomalies by calculating the distance between data points and their nearest neighbors. Outliers are typically points with the largest average distance to their neighbors

DISTANCE-BASED OUTLIER DETECTION

Distance-Based KNN

Dataset: 5 points in a 2D space: $A(2, 2)$, $B(2, 3)$, $C(3, 2)$, $D(10, 10)$, $E(2, 2.5)$

Goal: Identify outliers using $k = 2$.

Method: Euclidean Distance formula: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Step 1: Calculate Distance Matrix

- $d(A, B): \sqrt{(2 - 2)^2 + (3 - 2)^2} = \sqrt{0 + 1} = 1$
- $d(A, C): \sqrt{(3 - 2)^2 + (2 - 2)^2} = \sqrt{1 + 0} = 1$
- $d(A, E): \sqrt{(2 - 2)^2 + (2.5 - 2)^2} = \sqrt{0 + 0.25} = 0.5$
- $d(A, D): \sqrt{(10 - 2)^2 + (10 - 2)^2} = \sqrt{64 + 64} = \sqrt{128} \approx 11.3$

DISTANCE-BASED OUTLIER DETECTION

Distance-Based KNN

Step 2: Find $k = 2$ Nearest Neighbors (NN) and their Average Distance

A: Neighbors are E (0.5), B/C (1). $d(A, 2NN) = (0.5 + 1)/2 = 0.75$

B: Neighbors are E (0.5), A (1). $d(B, 2NN) = (0.5 + 1)/2 = 0.75$

C: Neighbors are A (1), E (1.11). $d(C, 2NN) = (1 + 1.11)/2 = 1.05$

D: Neighbors are C (≈ 9.9), A (≈ 11.3). $d(D, 2NN) = (9.9 + 11.3)/2 = 10.6$

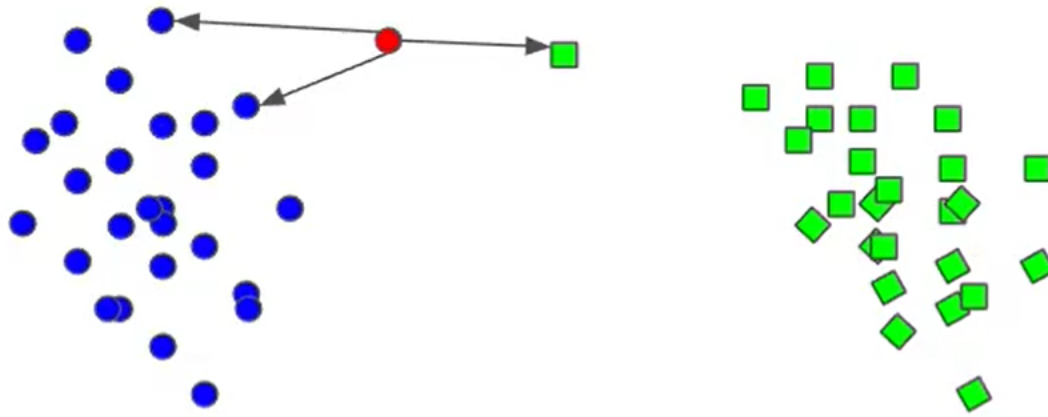
E: Neighbors are A (0.5), B (0.5). $d(E, 2NN) = (0.5 + 0.5)/2 = 0.5$

Step 3: Identify Outlier

Point **D** has the highest average distance (≈ 10.6) to its 2-nearest neighbors, making it the most significant outlier.

Distance from K Neighbours-Supervised

$K = 3$

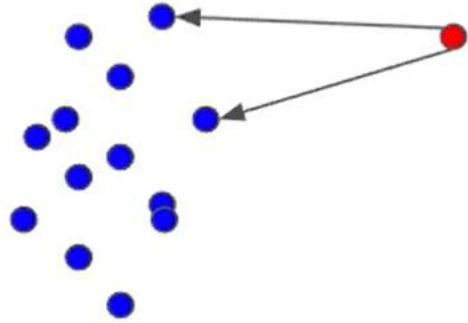


Blue neighbours = 2

Green neighbours = 1

Class = Blue

Distance from K Neighbours-**Unsupervised**

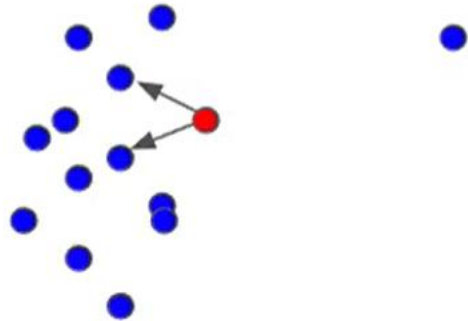


$K = 2$

1st neighbour = 4

2nd neighbour = 6

Avg = 5



1st neighbour = 0.9

2nd neighbour = 1

Avg = 0.95

Applications of KNN

KNN can be used for both classification and regression techniques but it is generally used for classification problems.

Examples of real-life use of KNN come in the fields of-

1. Microbiology(for classifying of cells),
2. Marketing(for customer segmentation)
3. Credit Fraud Analytics and many more...

DISTANCE-BASED OUTLIER DETECTION

Isolation Forest



DISTANCE-BASED OUTLIER DETECTION

Isolation Forests

Isolation forest is an **unsupervised learning** algorithm for outlier detection. The algorithm is based on Decision Trees.

The algorithm *isolates* the observations by selecting a feature randomly. It then **randomly chooses a split value between the maximum and minimum values** of the feature selected. This can be **represented by a decision tree structure**. The path length from root to leaf in such decision trees is a measure of normality of the point. This path length is equivalent to the number of splittings required to isolate a sample. Outliers are a minority as compared to other regular data points. In the feature space, they lie further away from the regular data points. Hence, using such kind of decision tree and random partitioning, these points should be identified closer to the root of the tree. This is because outliers have shorter average path length. When several such random decision trees are aggregated into a forest, they most likely produce shorter path lengths for outlier points.

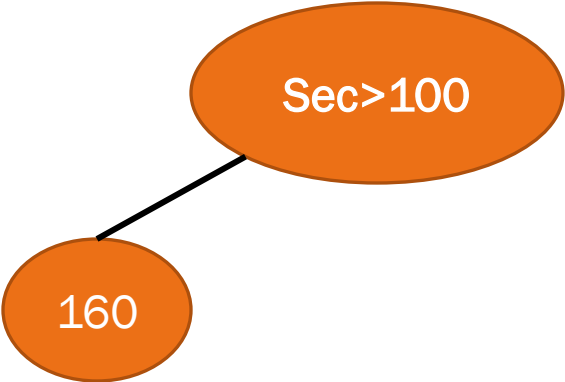
DISTANCE-BASED OUTLIER DETECTION

Name	Second they can hold breath
x	50
y	160
z	70
a	40
b	65
c	55

Sec>100

DISTANCE-BASED OUTLIER DETECTION

Name	Second they can hold breath
x	50
y	160
z	70
a	40
b	65
c	55



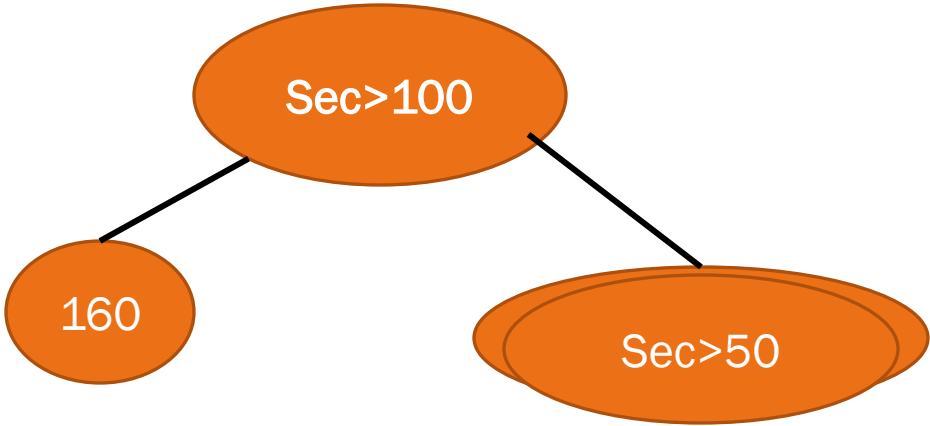
DISTANCE-BASED OUTLIER DETECTION

Name	Second they can hold breath
x	50
y	160
z	70
a	40
b	65
c	55



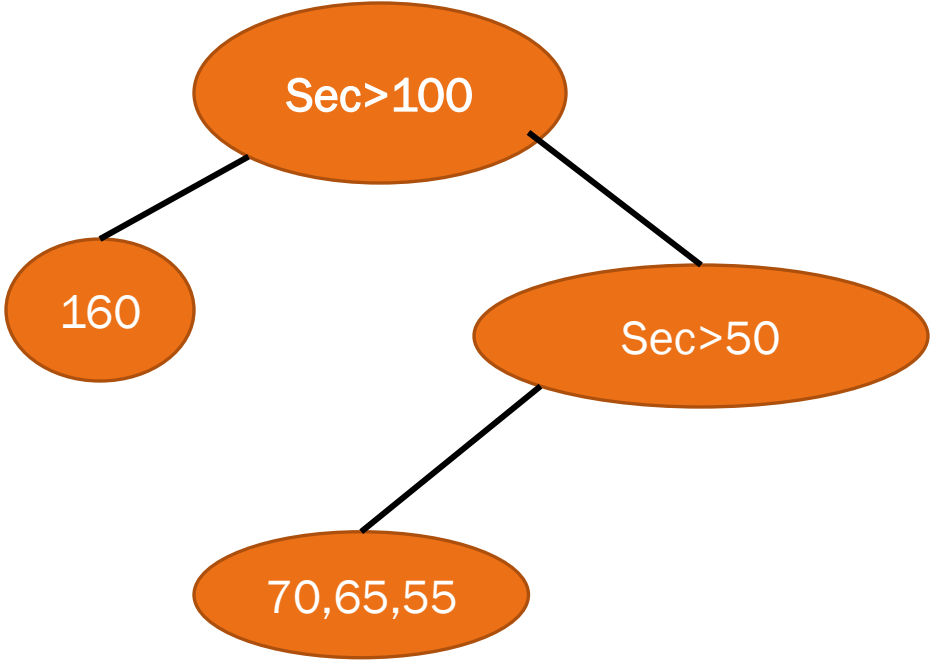
DISTANCE-BASED OUTLIER DETECTION

Name	Second they can hold breath
x	50
y	160
z	70
a	40
b	65
c	55



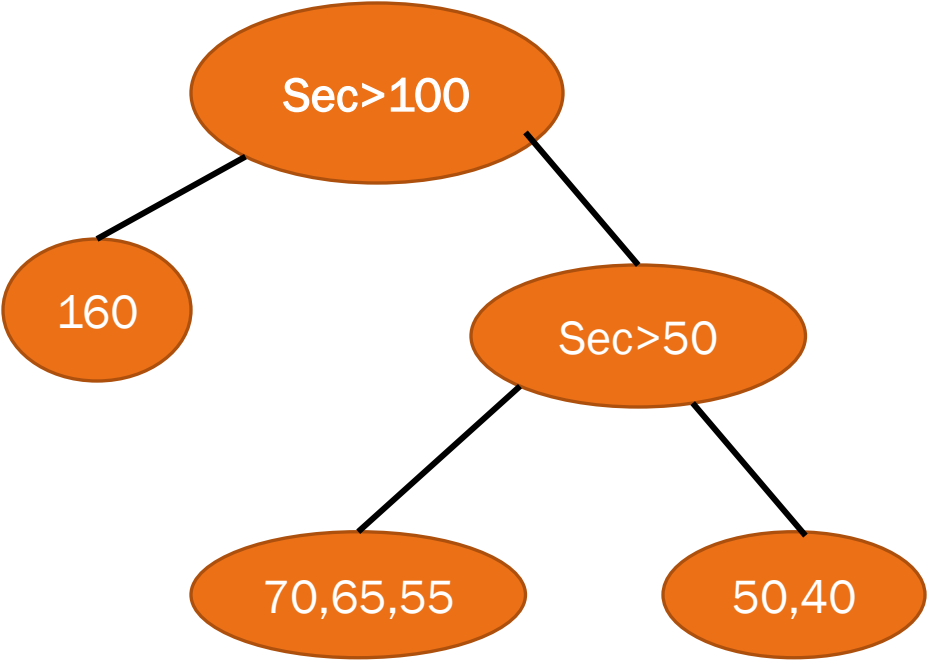
DISTANCE-BASED OUTLIER DETECTION

Name	Second they can hold breath
x	50
y	160
z	70
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b	65
c	55



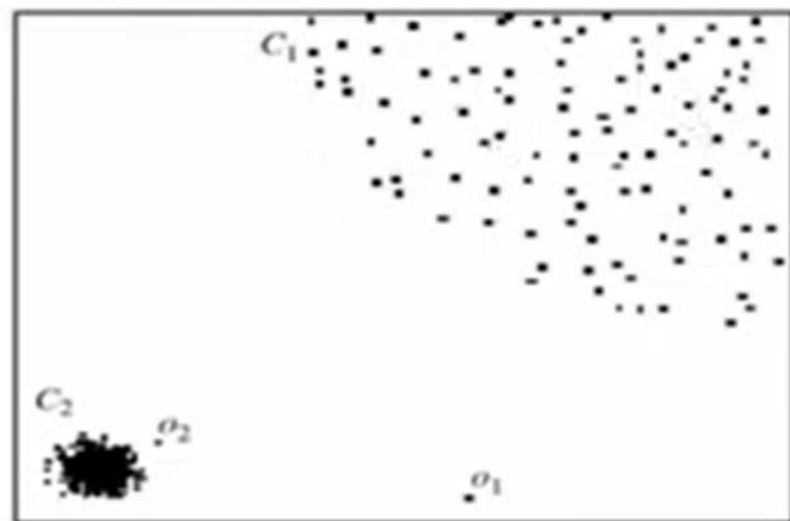
DISTANCE-BASED OUTLIER DETECTION

Name	Second they can hold breath
x	50
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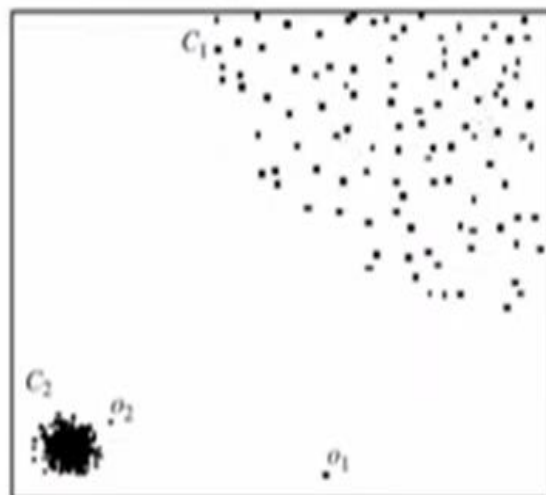


Density based Outlier detection

- Previous methods assume data are uniformly distributed
- Data may have **different density distributions**
 - Difficulty in choosing d_{min}



Density based Outlier detection

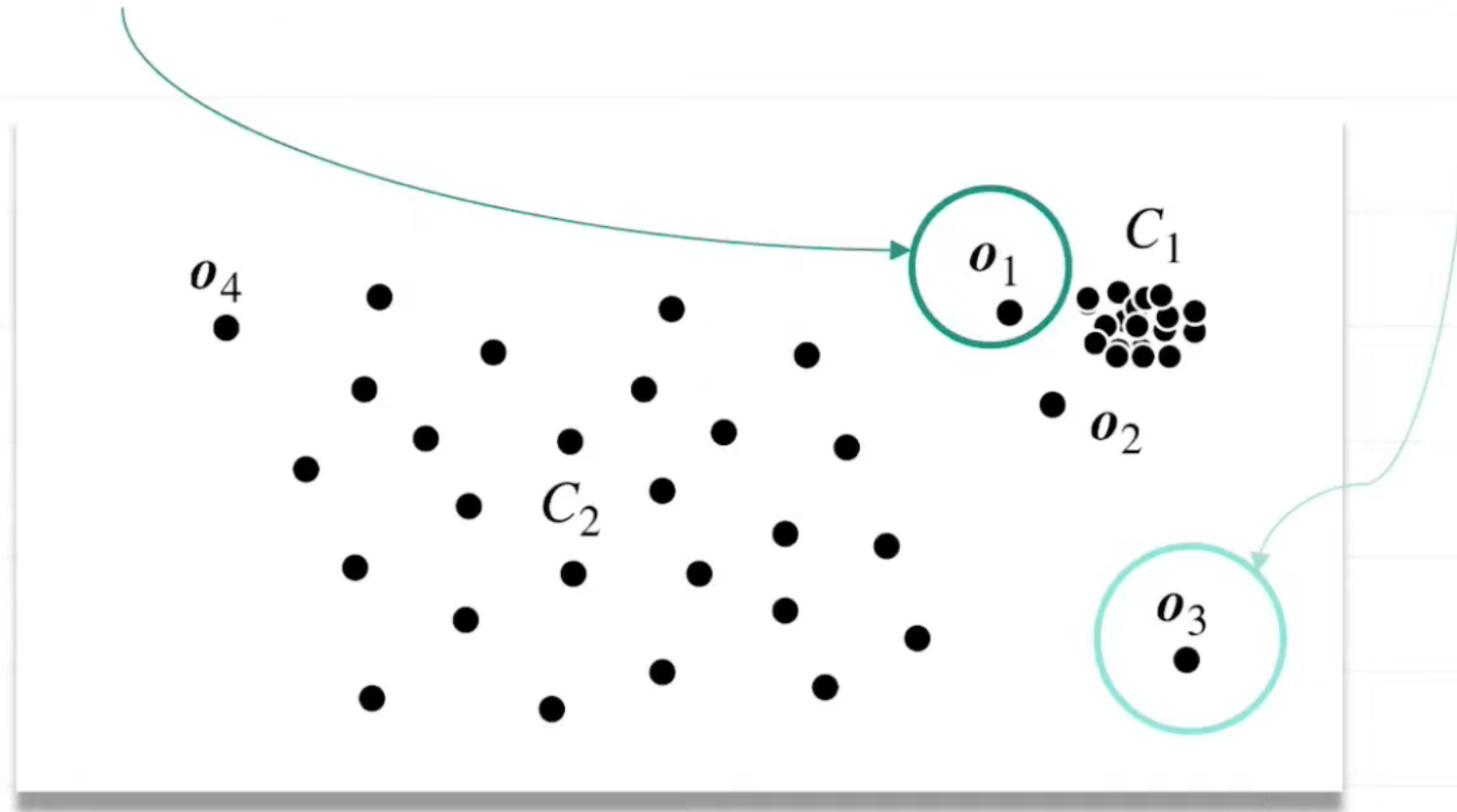


- **Local Outlier** – if its outlying relative to its local neighbourhood particularly wrt the density of the neighborhood
 - O2 is a local outlier wrt C2; o1 is also an outlier; none of the objects in C1 are treated as outliers
- Considers degree to which an object is an outlier
 - **Local Outlier factor** – degree depends on how isolated the object is wrt its surroundings

Types of outliers

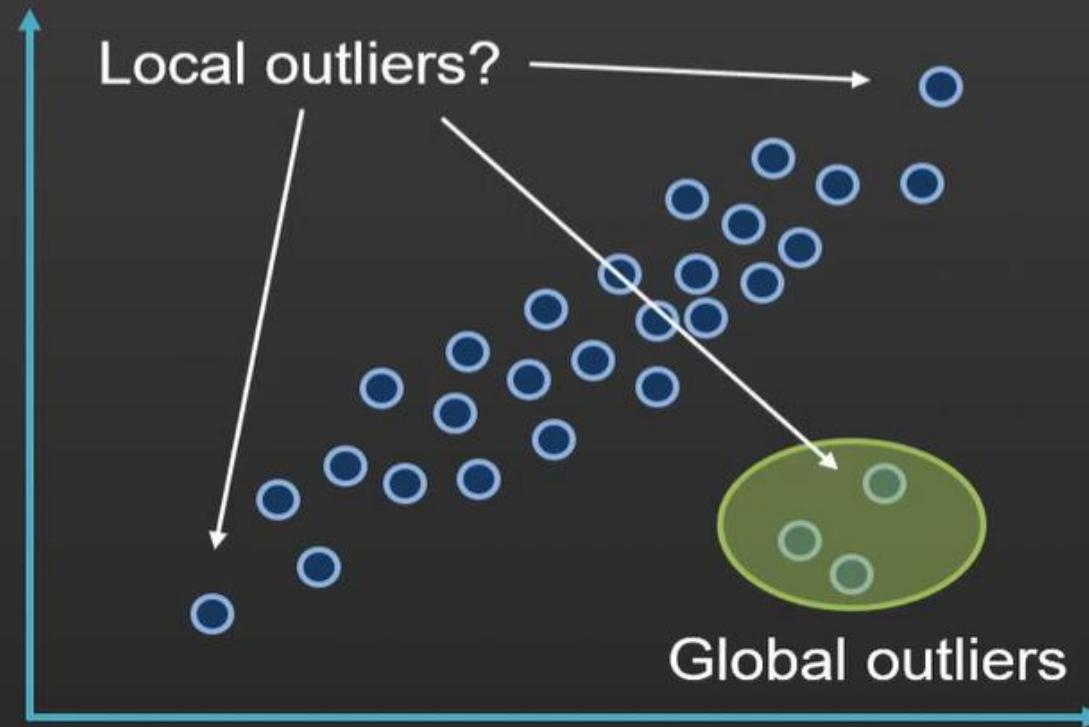
Local Outliers

Global Outliers



Global vs. Local Outliers

- k-NN great at detecting **global** outliers, but not **local** outliers.



Local Outlier Factor (LOF)

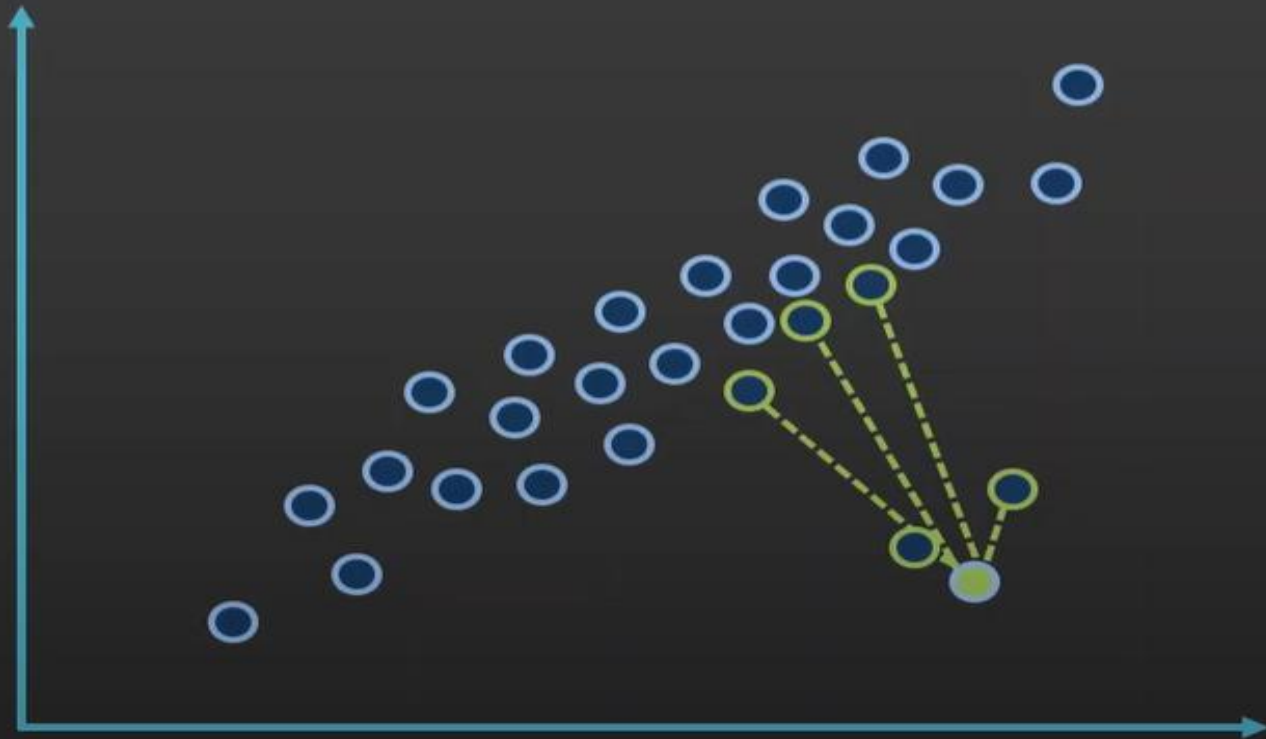
- Ratio (comparison) of the average **density** of the k-NN of an observation to the **density** of the observation itself.
 - > 1 means more likely to be anomaly
 - < 1 means less likely to be anomaly

Wait... What is Density?

- Inverse of the average **reachability** (distances) from observation to all its k-NN.
- Essentially, how far do we have to travel to nearest point, so less dense means farther travel.

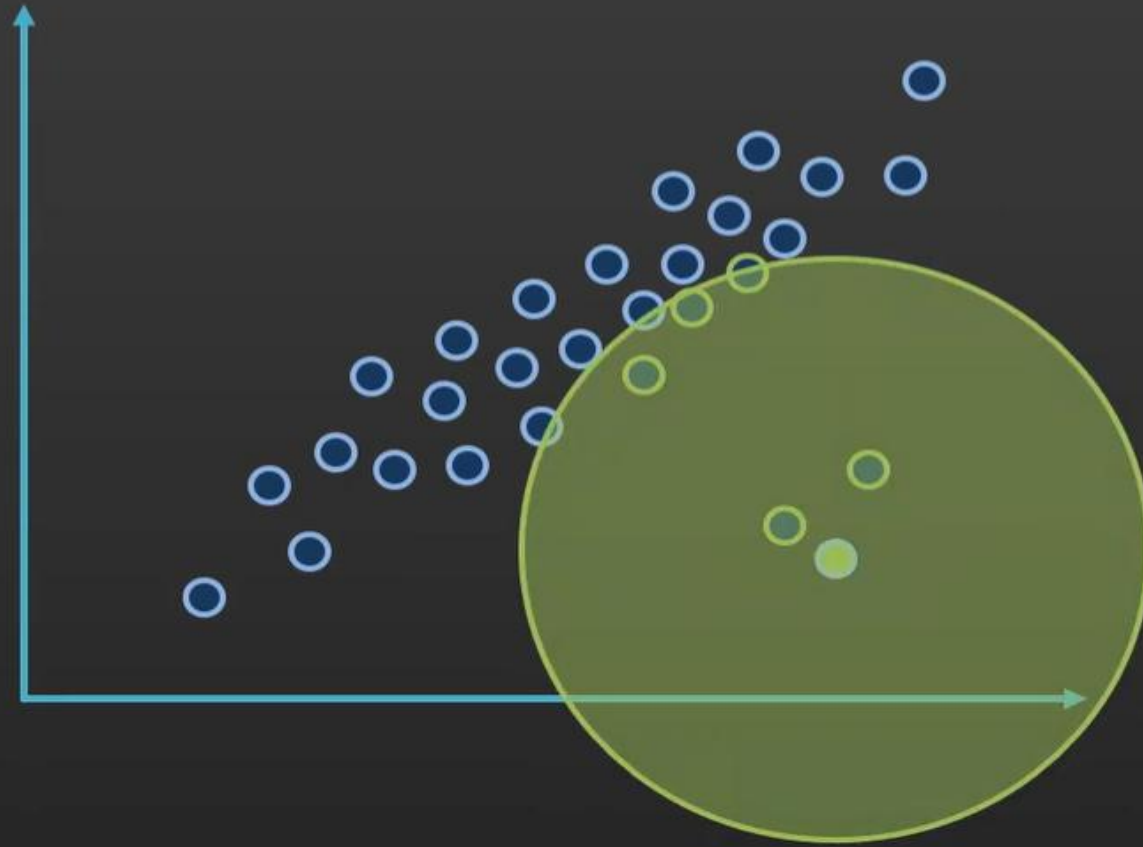
5-Nearest Neighbors Visually

- Need to measure distance to **5** nearest neighbors



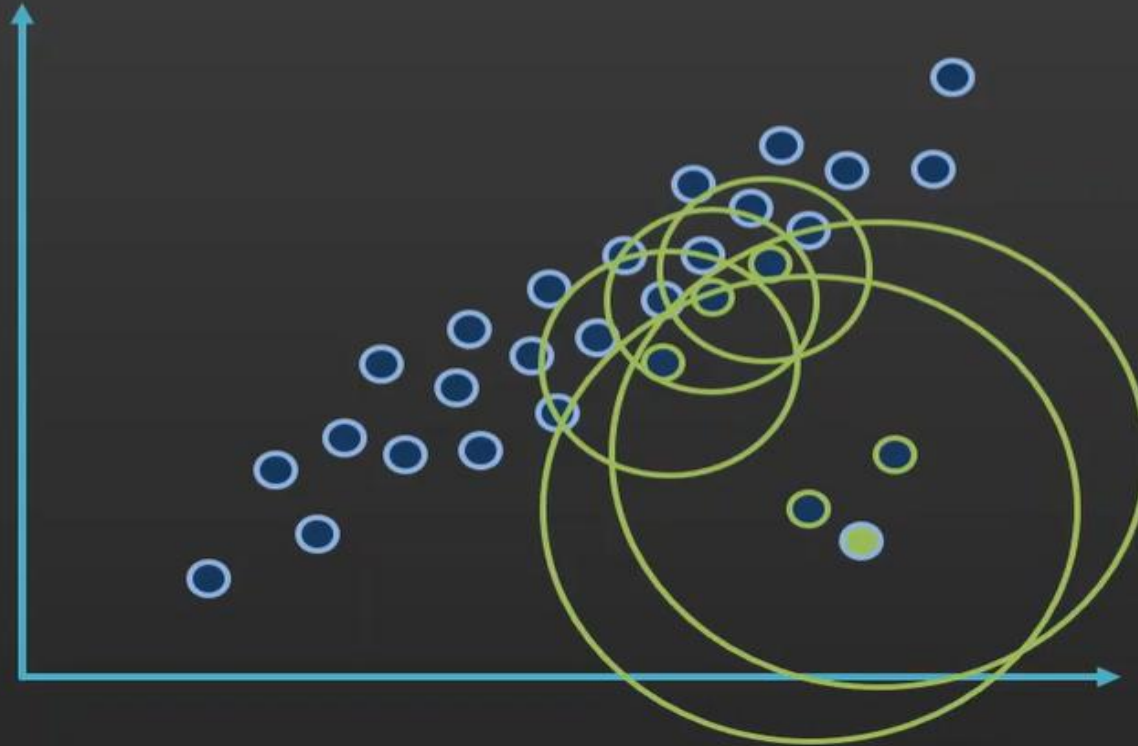
Density

- Essentially, how far do we have to travel to nearest point, so less dense means farther travel.



Local Outlier Factor (LOF)

- Ratio (comparison) of the average **density** of the k-NN of an observation to the **density** of the observation itself.

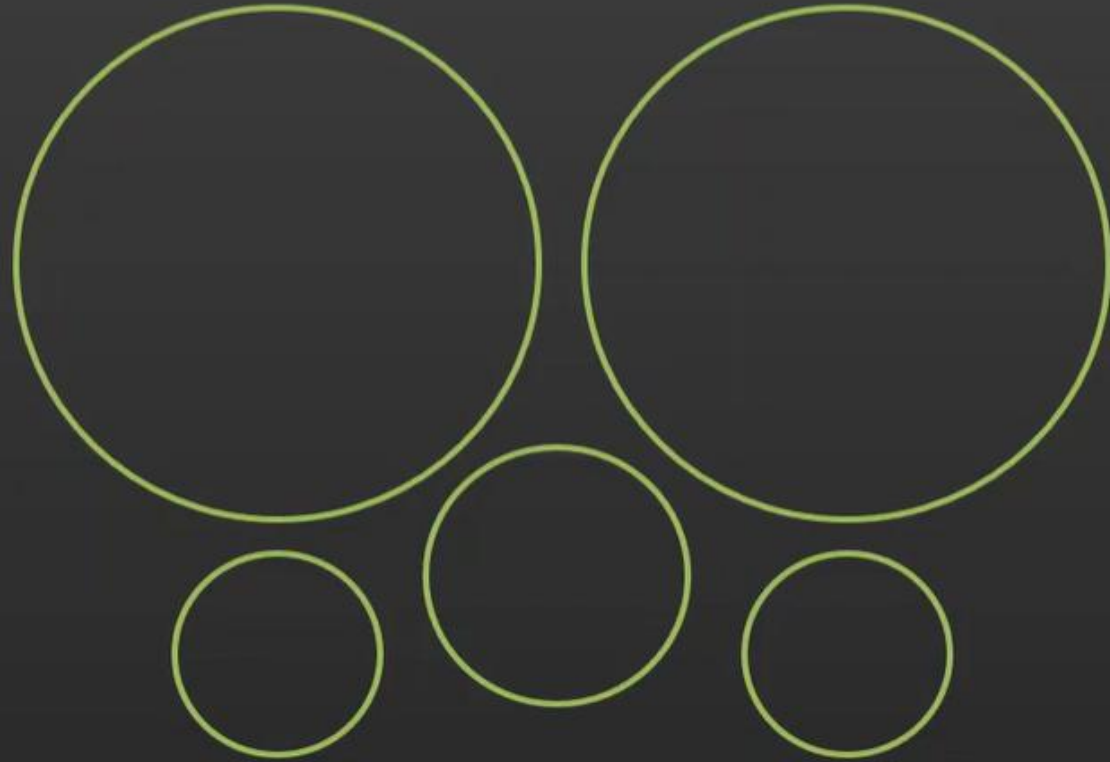


Time to Compare!

- Ratio (comparison) of the average **density** of the k-NN of an observation to the **density** of the observation itself.



vs.



Oh, I See it Now!

- Ratio (comparison) of the average **density** of the k-NN of an observation to the **density** of the observation itself.

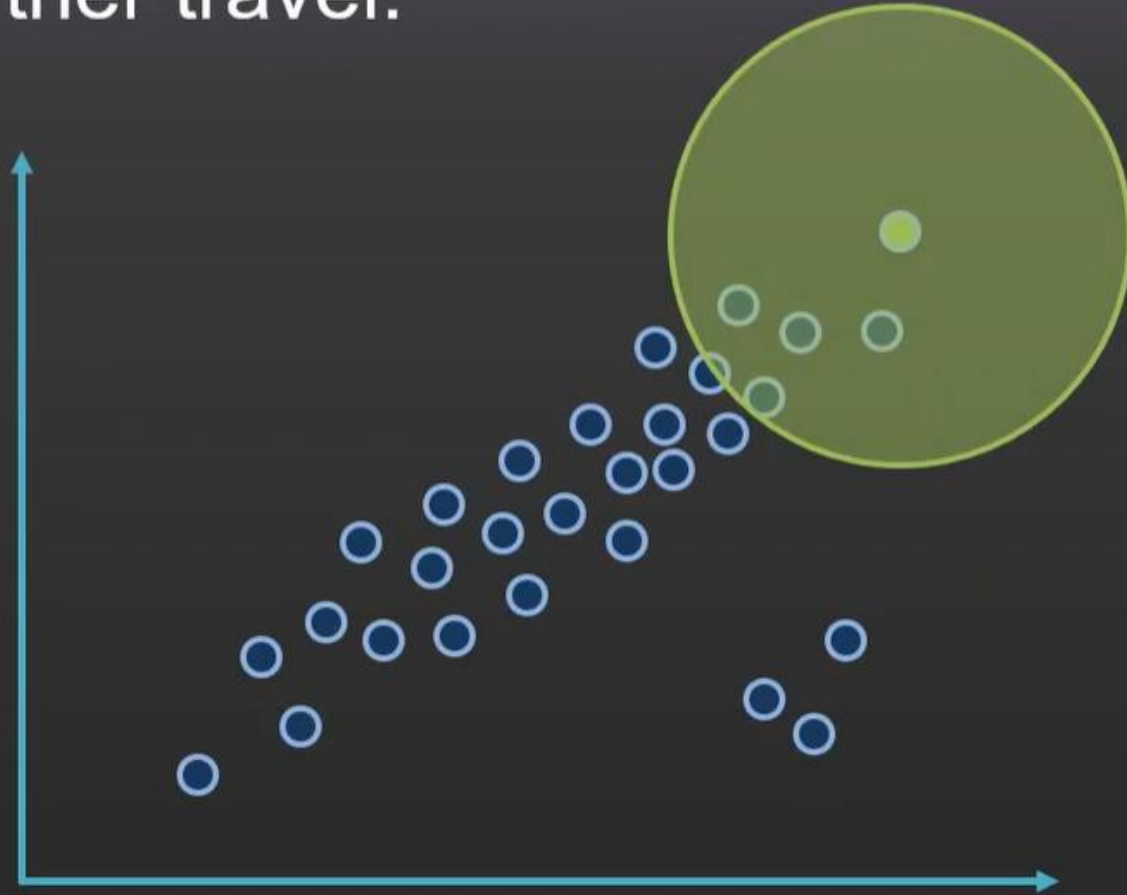


vs.



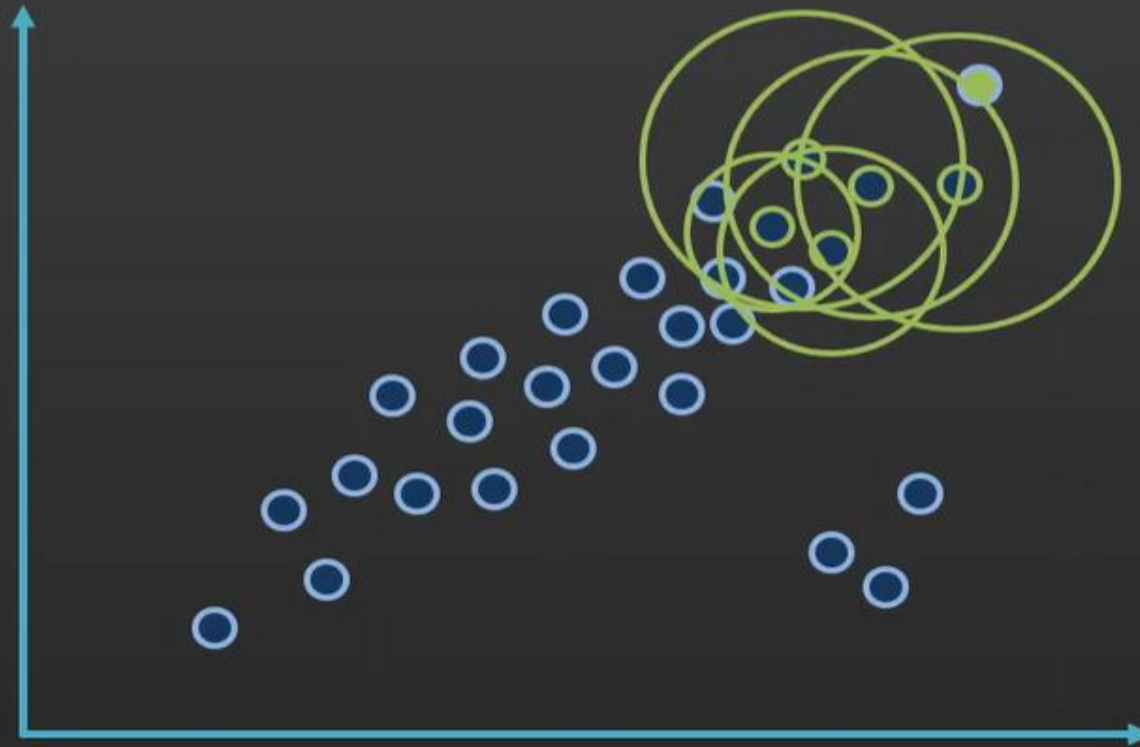
How About Another !?!

- Essentially, how far do we have to travel to nearest point, so less dense means farther travel.



How About Another !?!

- Ratio (comparison) of the average **density** of the k-NN of an observation to the **density** of the observation itself.



Time to Compare!

- Ratio (comparison) of the average **density** of the k-NN of an observation to the **density** of the observation itself.



vs.



Time to Compare!

- Ratio (comparison) of the average **density** of the k-NN of an observation to the **density** of the observation itself.

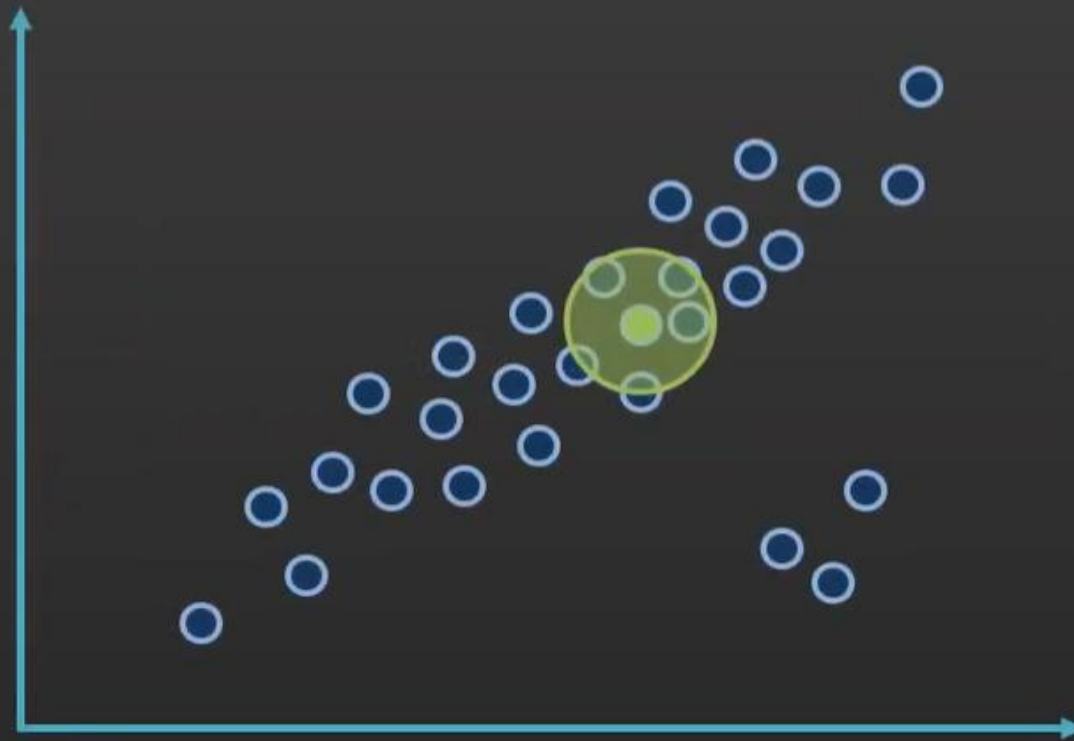


vs.



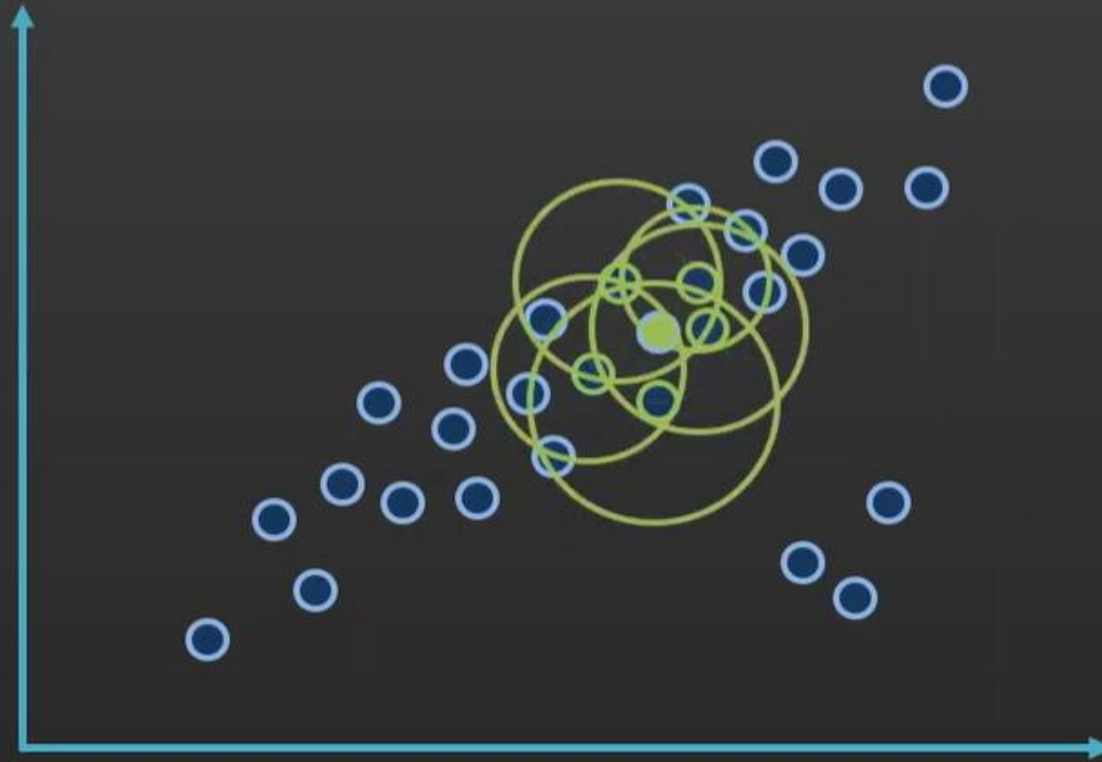
One More Time!

- Essentially, how far do we have to travel to nearest point, so less dense means farther travel.



Repeat!

- Ratio (comparison) of the average **density** of the k-NN of an observation to the **density** of the observation itself.



Time to Compare!

- Ratio (comparison) of the average **density** of the k-NN of an observation to the **density** of the observation itself.



vs.



Time to Compare!

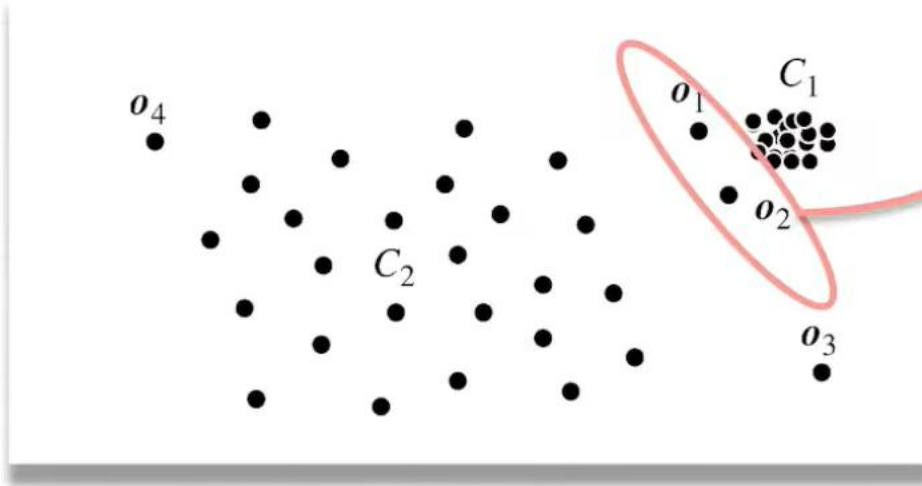
- Ratio (comparison) of the average **density** of the k-NN of an observation to the **density** of the observation itself.



vs.

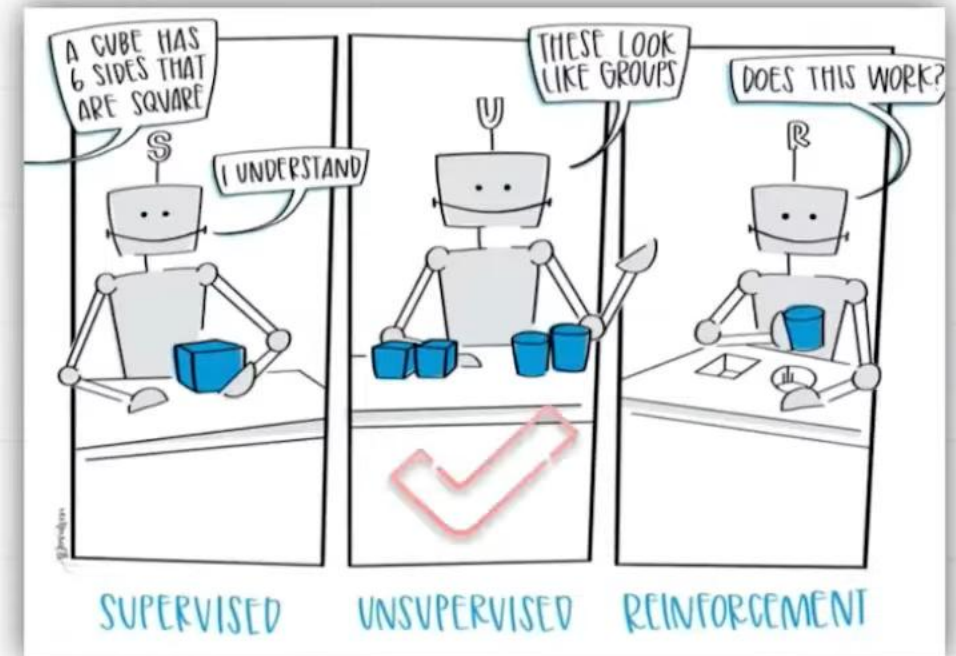


Local Outlier Factor

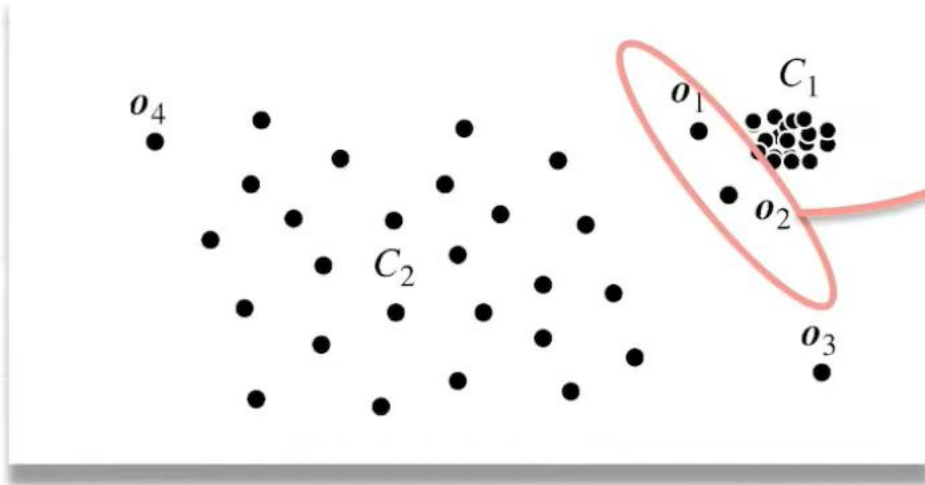


Local Outliers
Spotted!

Image Courtesy : Han, J., Kamber, M., & Pei, J. (2012). *Data mining: Concepts and techniques, third edition* (3rd ed.). Waltham, Mass.: Morgan Kaufmann Publishers.

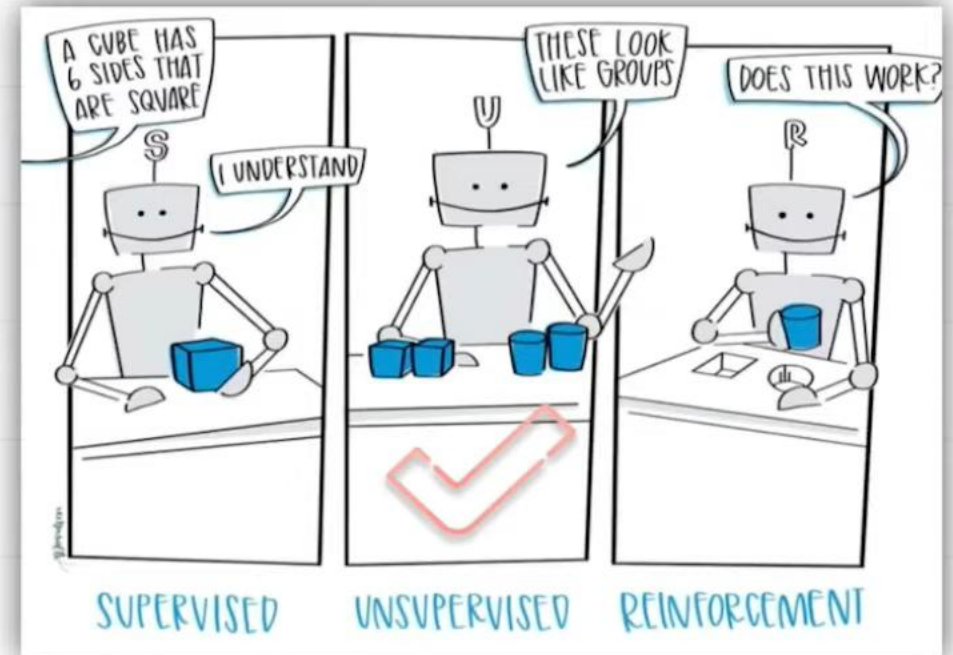
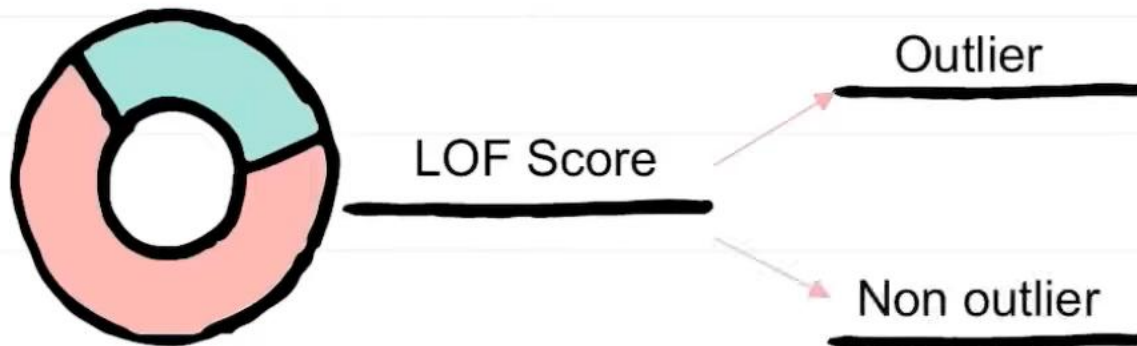


Local Outlier Factor



Local Outliers
Spotted!

Image Courtesy : Han, J., Kamber, M., & Pei, J. (2012). *Data mining: Concepts and techniques, third edition* (3rd ed.). Waltham, Mass.: Morgan Kaufmann Publishers.

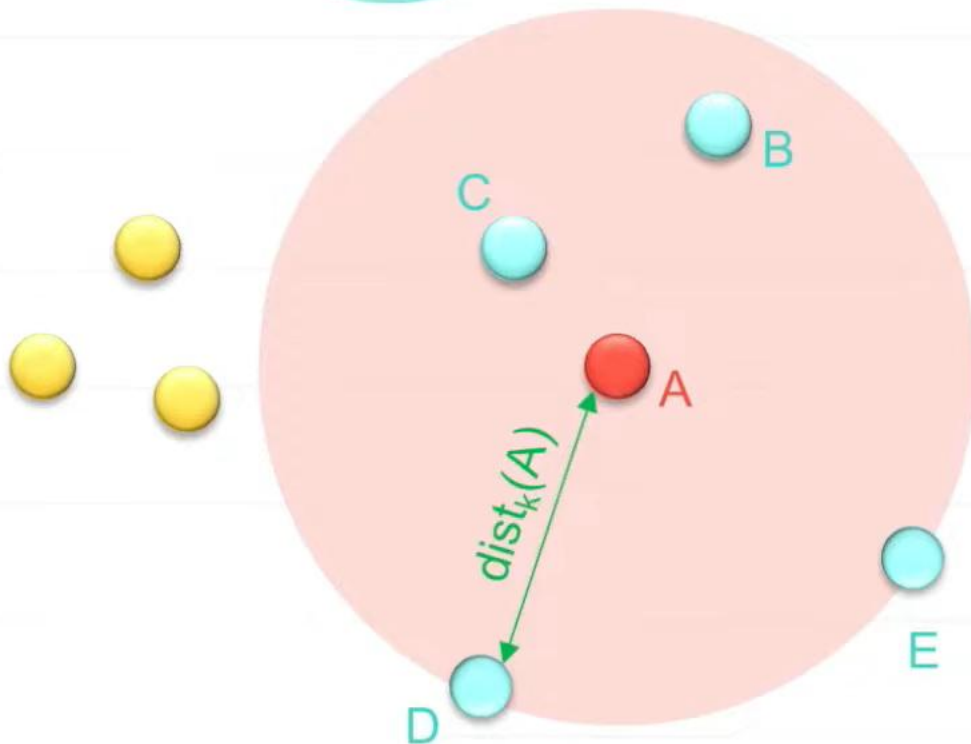


Terms to remember

If $k=3$:

$N_k(A) = \{B, C, D, E\}$

$\text{dist}_k(A) = \text{dist}(A, D)$



1. K-distance of A [$\text{dist}_k(A)$]:

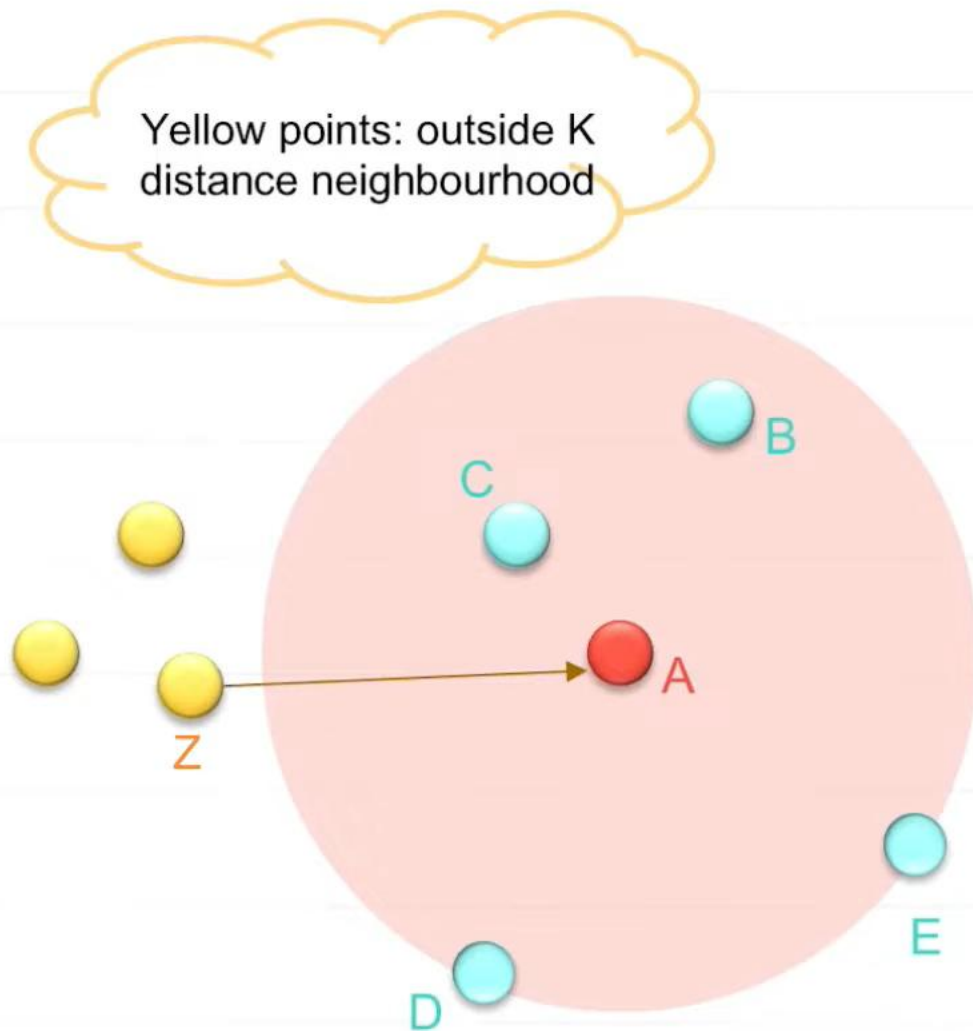
Distance between A and its k-nearest neighbour.

2. K-distance neighbourhood of A:

$$N_k(A) = \{A' \mid A' \in D, \text{dist}(A, A') \leq \text{dist}_k(A)\}$$



Terms to remember



3. Reachability distance from A' to A :

$$\text{reachdist}_k(A \leftarrow A') = \max\{\text{dist}_k(A), \text{dist}(A, A')\}$$

4. Local Reachability distance of A :

$$\text{lrd}_k(A) = \|N_k(A)\| / \sum(\text{reachdist}_k(A' \leftarrow A)), \text{ for all } A' \in N_k$$



Terms to remember

5. Local Outlier Factor score for A:

$LOF_k(A) = \frac{\sum(lrd_k(A'))}{(lrd(A) * \|N_k(A)\|)}$, for
all $A' \in N_k$

6. Threshold :

- Value greater than 1
- If $LOF_k(A) > \text{threshold}$, A is an outlier
- If $LOF_k(A) < 1$, A is not an outlier

ALGORITHM



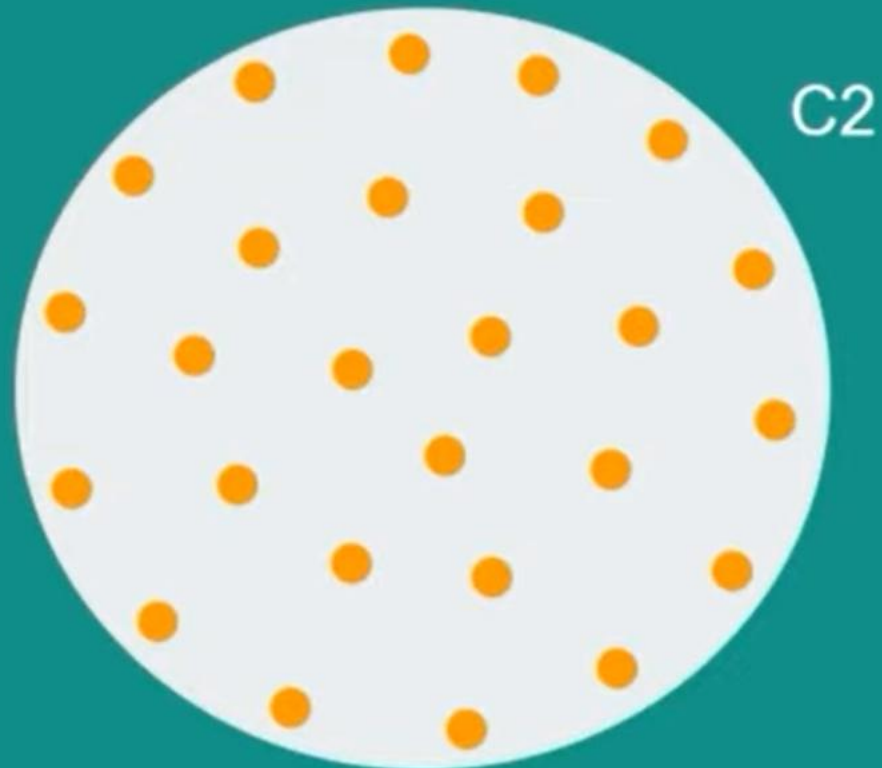
Algorithm

1. Calculate the reachability distance from all the neighbours (in the k -distance neighbourhood of the given point) to the given point.
2. Calculate the local reachability distance of the point.
3. Calculate the LOF score of the point using the local reachability of its neighbours and the point itself.
4. If $\text{LOF score} > \text{threshold}$, object is an outlier.
5. Repeat steps 1 through 4 for all points in the dataset.

C1 is dense cluster

C2 is sparse cluster

x1 & x2 are outliers

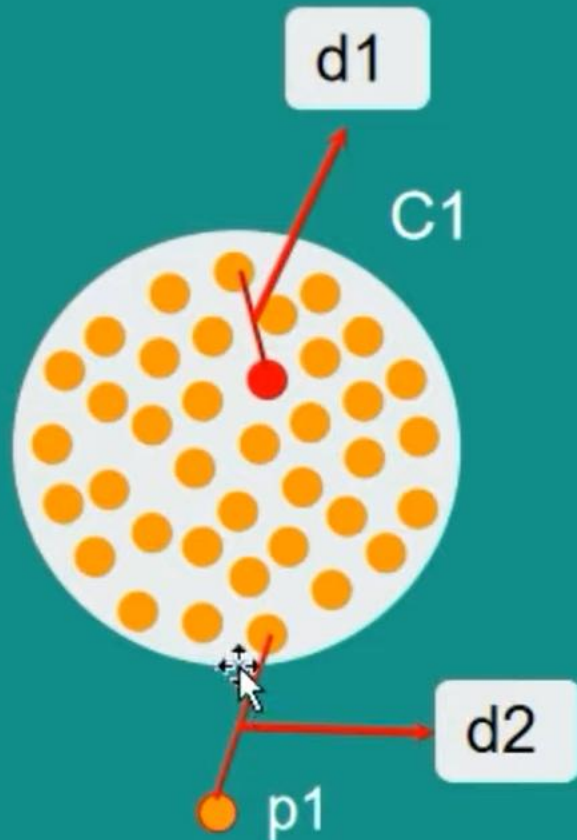


Outliers

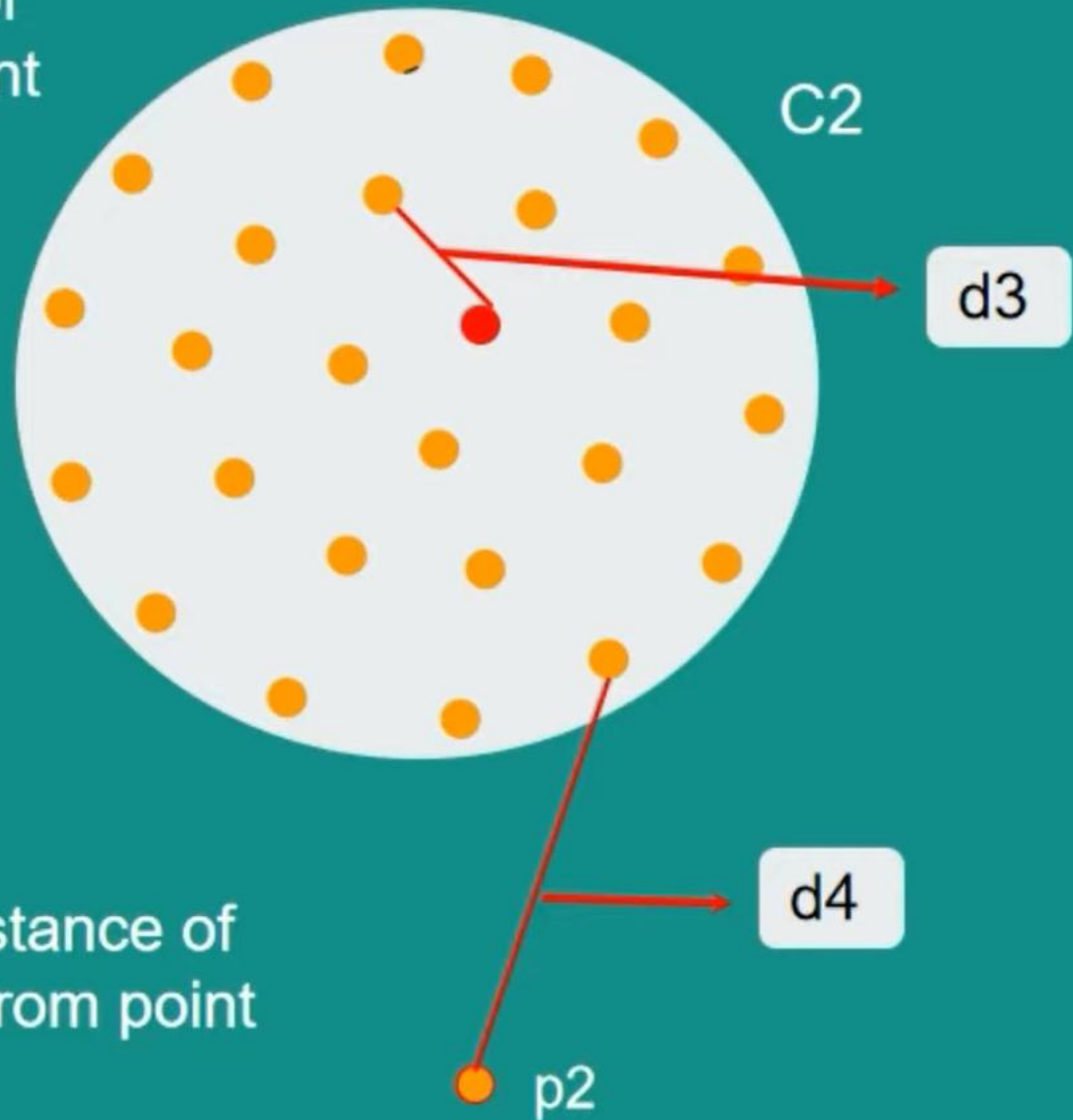


Let $k = 5$

d_2 is average distance of k nearest points from point p_1



d_4 is average distance of k nearest points from point p_2



A man with a beard and short dark hair, wearing a light blue button-down shirt, is sitting at a desk. He is looking towards the camera with a slight smile. His right hand is near his face, and his left hand is resting on a laptop. A glass of coffee is on the desk in front of him. The background is a bright, out-of-focus indoor setting.

**If you don't build your dream,
someone else will hire you
to help them build theirs.**

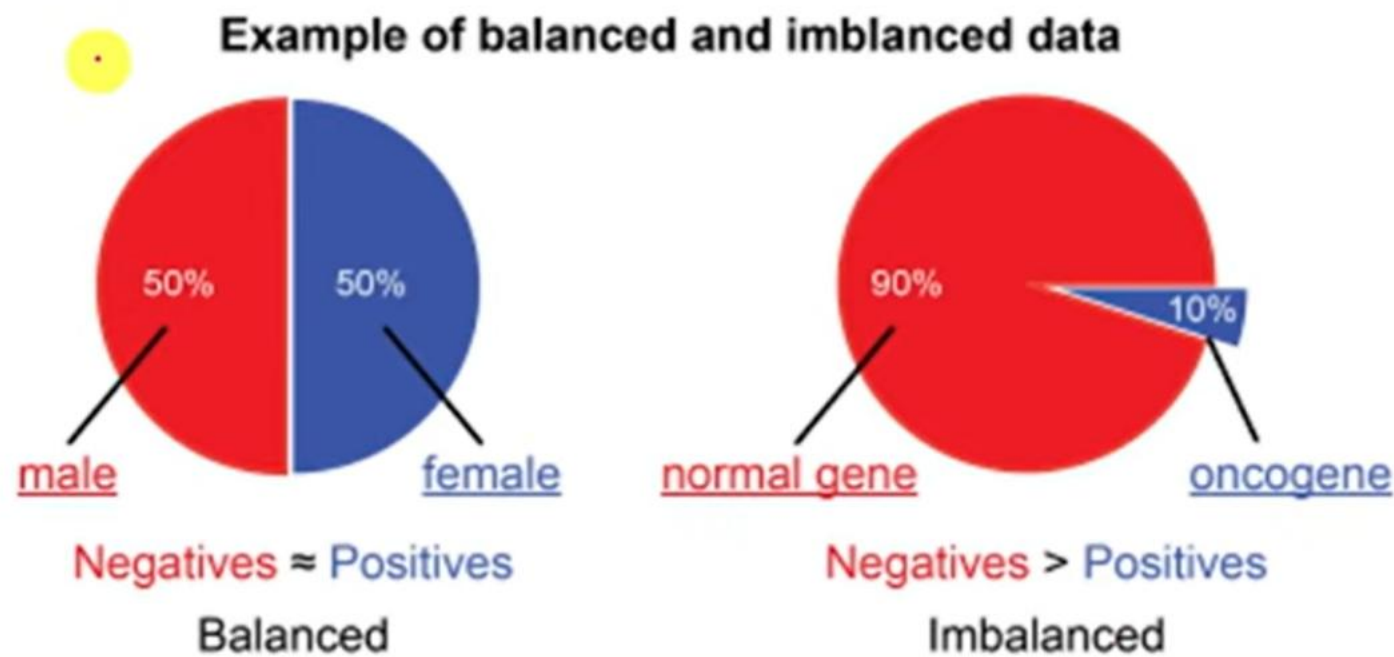
Dhirubhai Ambani

EVERYDAYPOWER

Module 4:-Anomaly Detection

- 4.1 Outliers,
Causes of Outliers,
Anomaly detection techniques,
Outlier Detection using Statistics
- 4.2 Outlier Detection using Distance based method,
Outlier detection using density-based methods,
SMOTE

How to handle Imbalanced Dataset



-
- **Balanced Dataset: —**
 - Let's take a simple example if in our data set, we have positive values which are approximately same as negative values.



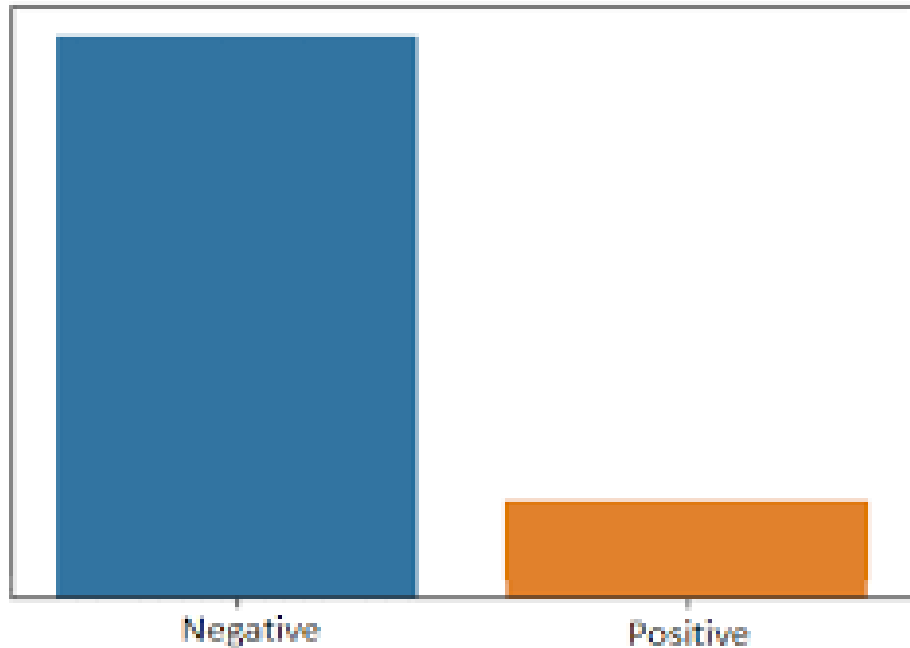
- **Imbalanced Dataset: —**
- If there is the very high difference between the positive values and negative values.



SMOTE

Imbalanced Dataset

An individual in the domain of Machine Learning is likely to come across a dataset where the class labels distribution is significantly different. In simple words, **Imbalanced Dataset usually reflects an unequal distribution of classes within a dataset.**



Problem with an Imbalanced Datasets

- Standard Machine Learning techniques such as Decision Tree and Logistic Regression have a bias towards the **majority** class, and they tend to ignore the minority class. ●

SMOTE

The term *accuracy can be highly misleading* as a performance metric for such data.

- Consider a dataset with 1000 data points having 950 points of **class 1** and 50 points of **class 0**.
- If we have a model which predicts all observations as 1, the accuracy in such case would be $950/1000 = 95\%$.

However, is it really a good model?

SMOTE

we should analyze the **Recall, Precision and F1-scores** depending on the business requirements. Consider an example of cancer detection; here, False Negatives are of primary concern. The value of false negatives should be as low as possible since we do not want our model to predict a cancerous patient as non-cancerous.

Accuracy = (TP + TN)/(TP + TN + FP + FN)

Precision = TP / TP + FP

Recall = TP / TP + FN

F-measure = 2*((precision*recall)/(precision+recall))

Anomaly Detection Applications: Detection of Breast Cancer		
Output of Algorithm	Actual Normal	Actual Abnormal
Flagged as Normal	No Problem True Negative	Missed Alarm The patient dies  False Negative
Flagged as Abnormal	False Alarm - Require re-doing test, and follow-ups - Cause fear and worry  False positive	Victory True Positive

SMOTE

If there is a dataset consisting of 10000 genuine and 10 fraudulent transactions, **the classifier will tend to classify fraudulent transactions as genuine transactions**. The reason can be easily explained by the numbers.

Suppose the machine learning algorithm has two possibly outputs as follows:

1. Model 1 classified 7 out of 10 fraudulent transactions as genuine transactions and 10 out of 10000 genuine transactions as fraudulent transactions.
2. Model 2 classified 2 out of 10 fraudulent transactions as genuine transactions and 100 out of 10000 genuine transactions as fraudulent transactions.

SMOTE

Name	Formula	Explanation
True Positive Rate (TP rate)	$TP / (TP + FP)$	The closer to 1, the better. TP rate = 1 when FP = 0. (No false positives)
True Negative Rate (TN rate)	$TN / (TN + FN)$	The closer to 1, the better. TN rate = 1 when FN = 0. (No false negatives)
False Positive Rate (FP rate)	$FP / (FP + TN)$	The closer to 0, the better. FP rate = 0 when FP = 0. (No false positives)
False Negative Rate (FN rate)	$FN / (FN + TP)$	The closer to 0, the better. FN rate = 0 when FN = 0. (No false negatives)

True Positive (TP) – An example that is **positive** and is classified correctly as **positive**

True Negative (TN) – An example that is **negative** and is classified correctly as **negative**

False Positive (FP) – An example that is **negative** but is classified wrongly as **positive**

False Negative (FN) – An example that is **positive** but is classified wrongly as **negative**

SMOTE

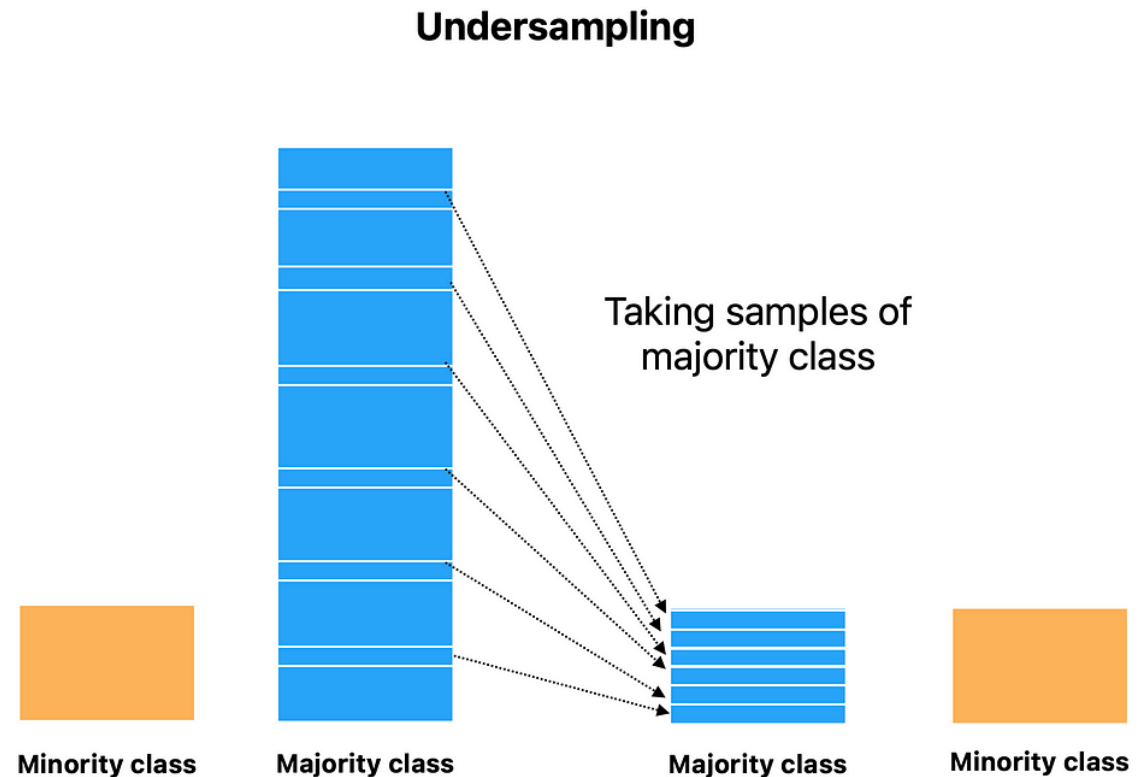
Sampling based approaches

This can be roughly classified into three categories:

1. Oversampling, by adding more of the minority class so it has more effect on the machine learning algorithm
2. Undersampling, by removing some of the majority class so it has less effect on the machine learning algorithm
3. Hybrid, a mix of oversampling and undersampling

SMOTE

Under-sampling randomly removes observations of the majority class. This reduces the number of majority class observations used in the training set and as a result balances the number of observations of the two classes better. This is suitable when you have a lots of observations in your dataset (>10K observations). **The risk is you are losing information and so may lead to underfitting.**

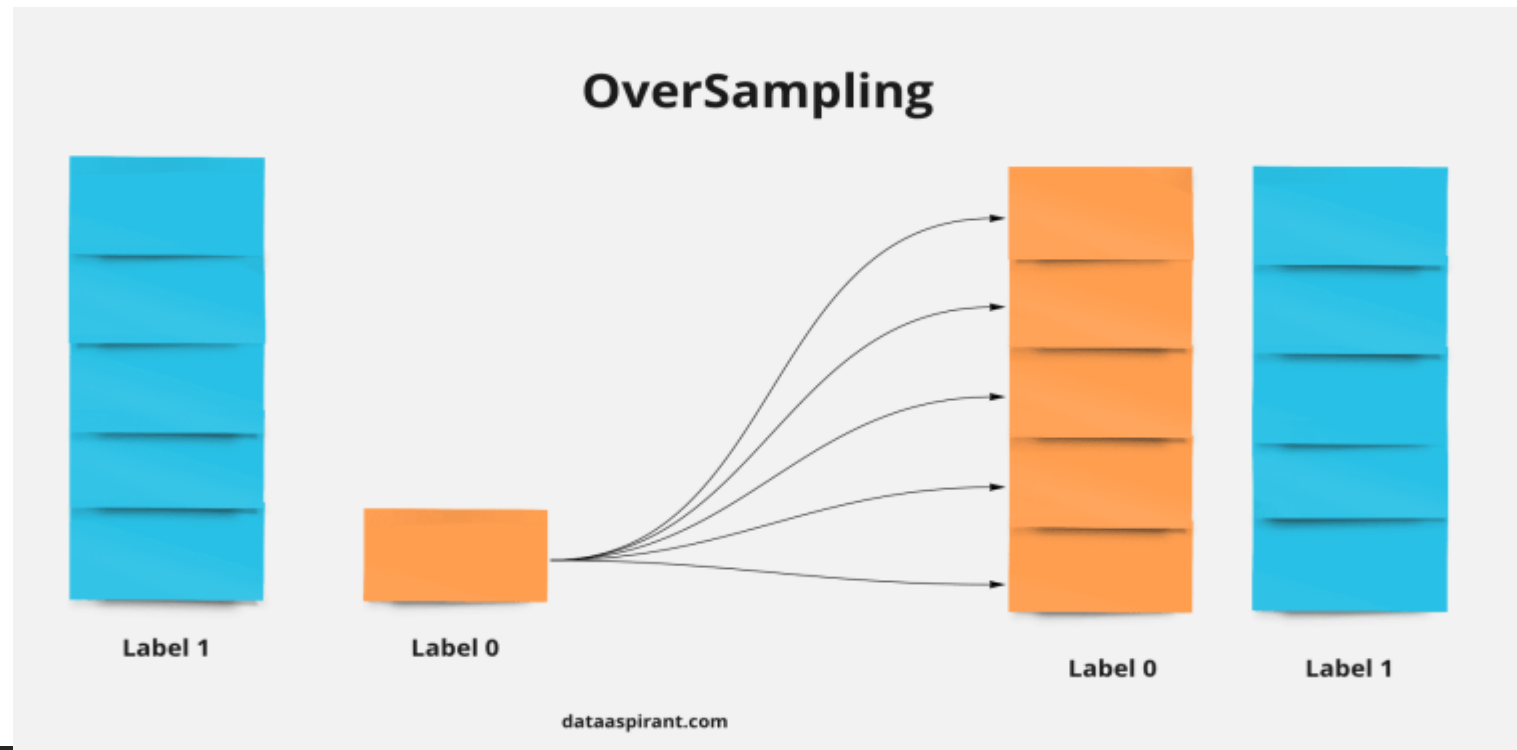


SMOTE

Oversampling (Upsampling) the minority class

Oversampling randomly duplicates observations from the minority class in order to make its signal stronger. The simplest form of oversampling is **sampling with replacement**.

Oversampling is suitable when you don't have a lots of observations in your dataset (<10K observations). **The risk is if you duplicate too many observations, well then you are overfitting.**



SMOTE

SMOTE: a powerful solution for imbalanced data

*SMOTE stands for **Synthetic Minority Oversampling Technique**. SMOTE is an improved method of dealing with imbalanced data in classification problems.*

SMOTE

Algorithm 1 A description of SMOTE over-sampling algorithm developed by Chawla et al (2002).

repeat

 Select a minority sample x_0 at random.

 Identify the K -nearest neighbors of x_0 among the minority samples.

 Randomly choose one of the K nearest neighbors of x_0 , from the previous step and denote the chosen sample as x_k where k is the rank of the chosen neighbor.

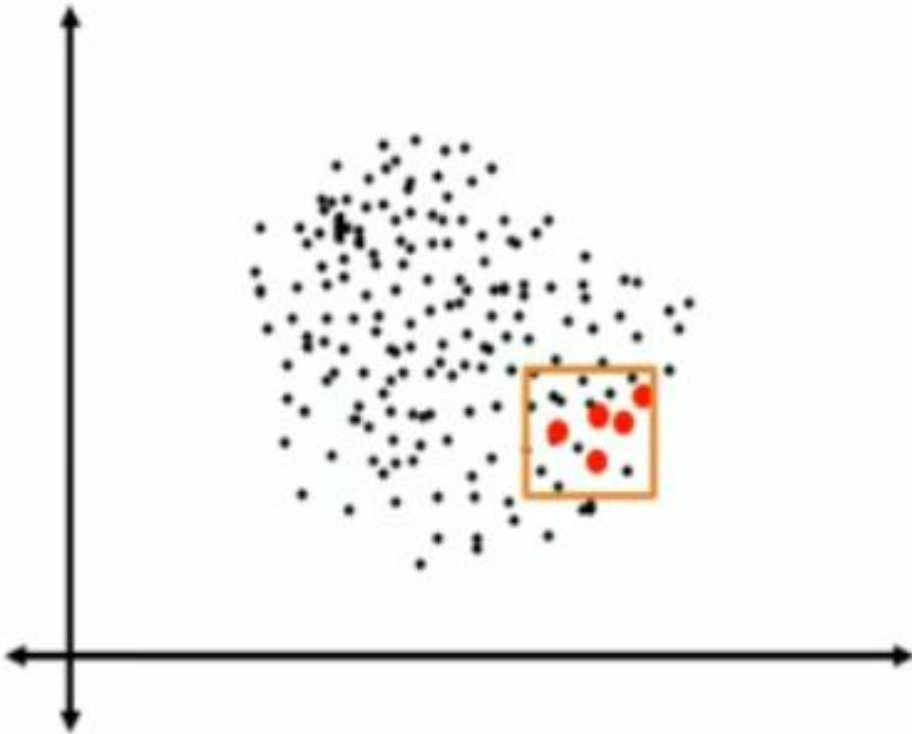
 Perform linear interpolation between x_0 and the chosen neighbor x_k to create a new synthetic sample z according to:

$z = x_0 + w(x_k - x_0)$ where w is a uniform random variable in the range $[0, 1]$.

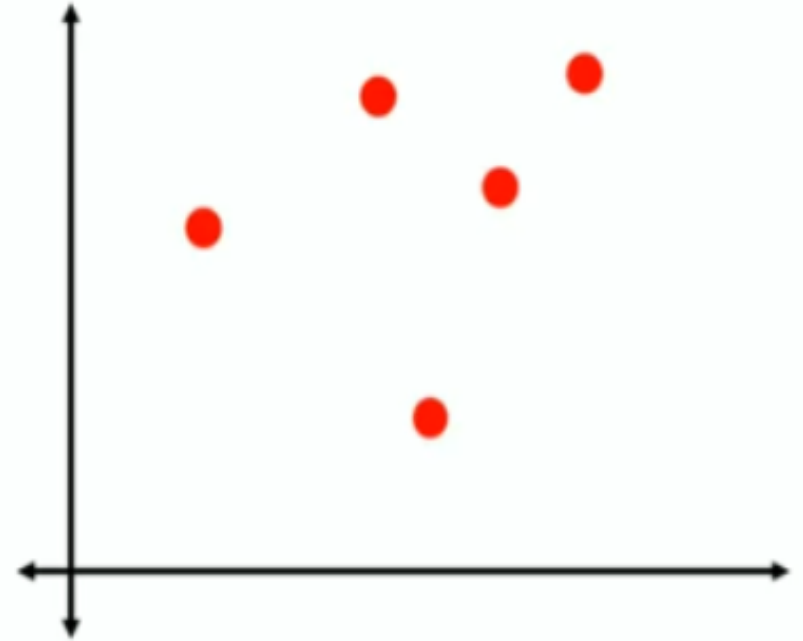
until M synthetic samples are generated.

SMOTE

Initial distribution

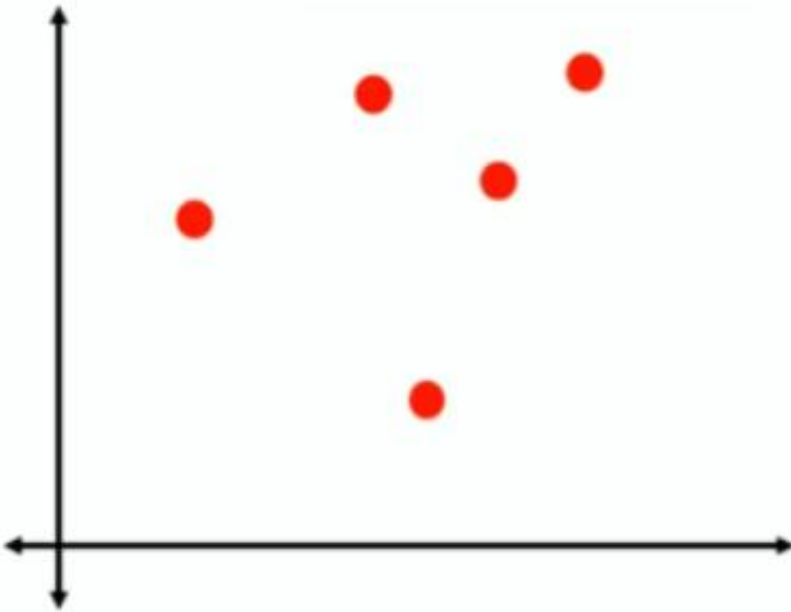


Considering only minority samples

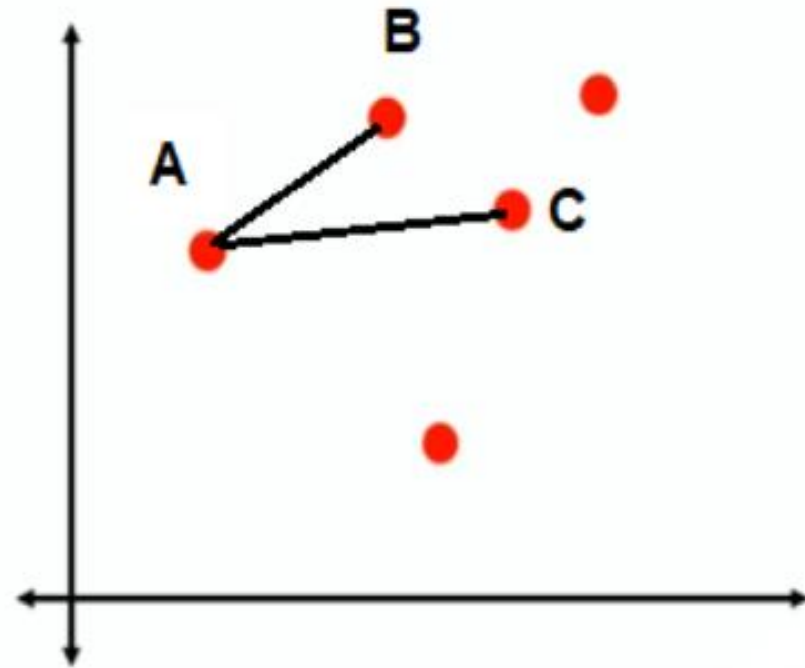


SMOTE

Considering only
minority samples

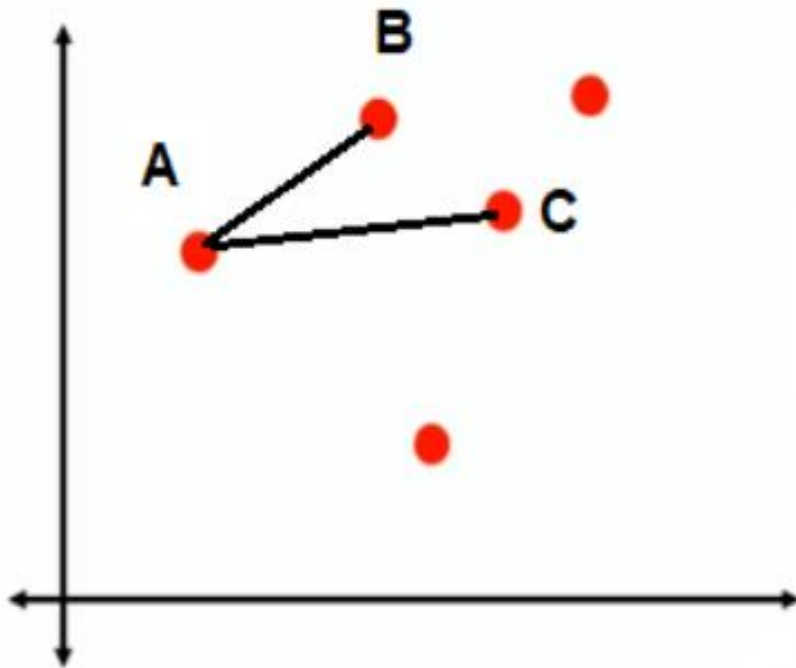


2 nearest neighbours of point A

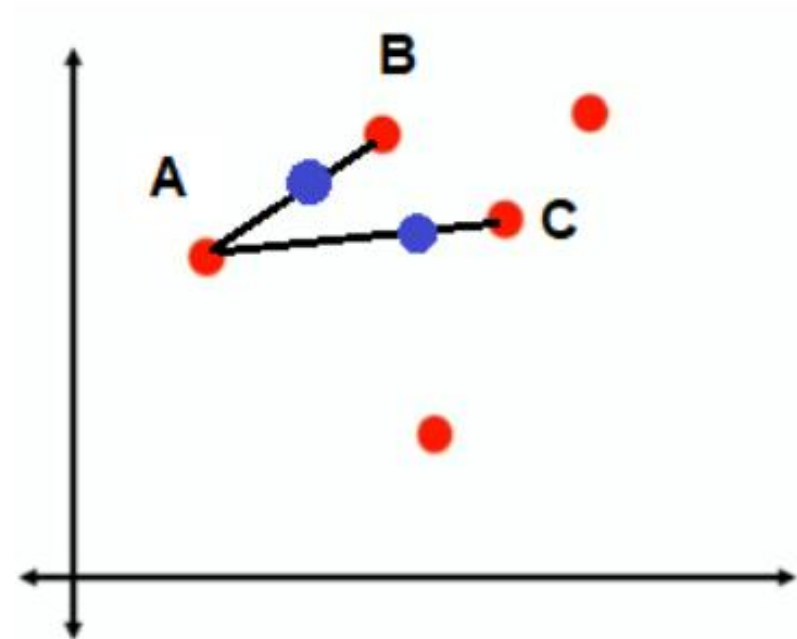


SMOTE

2 nearest neighbours of point A

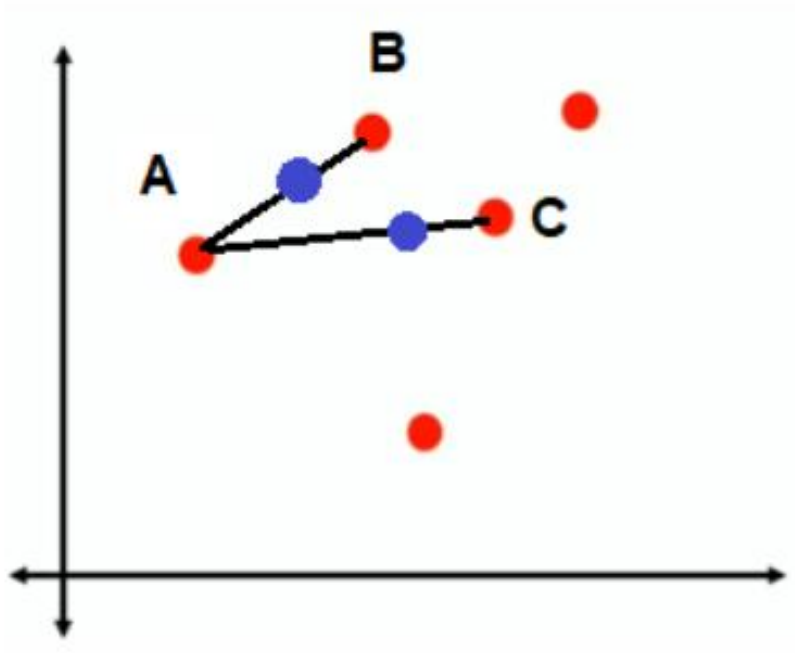


● Synthetic points generated between A and its 2 nearest neighbours

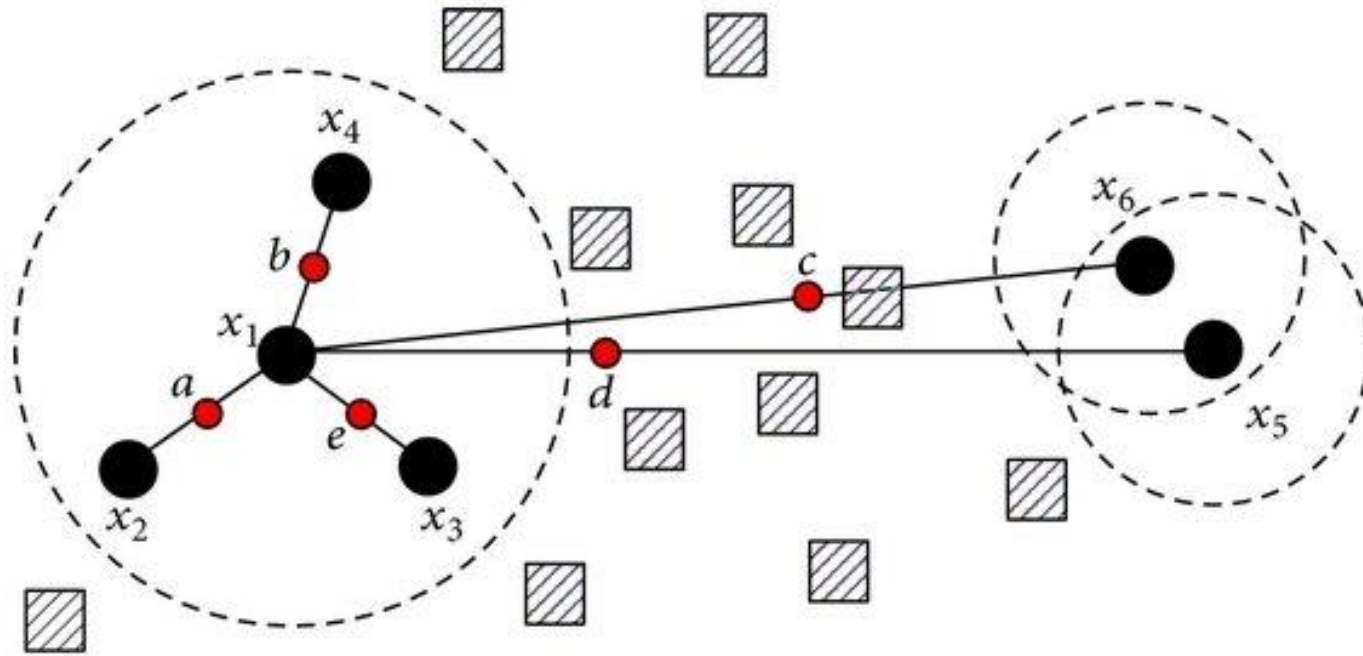


SMOTE

- Synthetic points generated between A and its 2 nearest neighbours



SMOTE



- Majority class samples
- Minority class samples
- Synthetic samples

Classification Report:

	precision	recall	f1-score	support
0.0	0.82	0.90	0.86	1033
1.0	0.63	0.45	0.53	374
accuracy			0.78	1407
macro avg	0.73	0.68	0.69	1407
weighted avg	0.77	0.78	0.77	1407

Under sampling

Classification Report:				
	precision	recall	f1-score	support
0	0.71	0.74	0.72	374
1	0.73	0.69	0.71	374
accuracy			0.72	748
macro avg	0.72	0.72	0.72	748
weighted avg	0.72	0.72	0.72	748

Classification Report:

	precision	recall	f1-score	support
0.0	0.82	0.90	0.86	1033
1.0	0.63	0.45	0.53	374
accuracy			0.78	1407
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Under sampling

Classification Report:

	precision	recall	f1-score	support
0	0.71	0.74	0.72	374
1	0.73	0.69	0.71	374
accuracy			0.72	748
macro avg	0.72	0.72	0.72	748
weighted avg	0.72	0.72	0.72	748

Classification Report:

	precision	recall	f1-score	support
0.0	0.82	0.90	0.86	1033
1.0	0.63	0.45	0.53	374
accuracy			0.78	1407
macro avg	0.73	0.68	0.69	1407
weighted avg	0.77	0.78	0.77	1407

Method2: Oversampling

Classification Report:					
	precision	recall	f1-score	support	
0	0.82	0.72	0.76	1033	
1	0.75	0.84	0.79	1033	
accuracy			0.78	2066	
macro avg	0.78	0.78	0.78	2066	I
weighted avg	0.78	0.78	0.78	2066	

Classification Report:

	precision	recall	f1-score	support
0.0	0.82	0.90	0.86	1033
1.0	0.63	0.45	0.53	374
accuracy			0.78	1407
macro avg	0.73	0.68	0.69	1407
weighted avg	0.77	0.78	0.77	1407

Method2: Oversampling

Classification Report:

	precision	recall	f1-score	support
0	0.82	0.72	0.76	1033
1	0.75	0.84	0.79	1033
accuracy			0.78	2066
macro avg	0.78	0.78	0.78	2066
weighted avg	0.78	0.78	0.78	2066

Classification Report:

	precision	recall	f1-score	support
0.0	0.82	0.90	0.86	1033
1.0	0.63	0.45	0.53	374
accuracy			0.78	1407
macro avg	0.73	0.68	0.69	1407
weighted avg	0.77	0.78	0.77	1407

SMOTE

```
Classification Report:
              precision    recall  f1-score   support

     0           0.81       0.81       0.81       1033
     1           0.81       0.81       0.81       1033

 accuracy              0.81       2066
 macro avg           0.81       0.81       0.81       2066
weighted avg           0.81       0.81       0.81       2066
```

Classification Report:

	precision	recall	f1-score	support
0.0	0.82	0.90	0.86	1033
1.0	0.63	0.45	0.53	374
accuracy			0.78	1407
macro avg	0.73	0.68	0.69	1407
weighted avg	0.77	0.78	0.77	1407

SMOTE

Classification Report:

	precision	recall	f1-score	support
0	0.81	0.81	0.81	1033
1	0.81	0.81	0.81	1033
accuracy			0.81	2066
macro avg	0.81	0.81	0.81	2066
weighted avg	0.81	0.81	0.81	2066

Module 4:-Anomaly Detection

- 4.1 Outliers,
Causes of Outliers,
Anomaly detection techniques,
Outlier Detection using Statistics
- 4.2 Outlier Detection using Distance based method,
Outlier detection using density-based methods,
SMOTE

Be
Good
Do
Good

**To be good, and to
do good, is all
we have to do.**
